

SMITHIAN FOLD THEORY V3

From One Relation to Society

An Exact, Zero-Parameter and Machine-Closed Reconstruction of Social and Collective Sciences
from Smithian Fold Theory

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Contents

Contents	2
An Exact, Zero-Parameter and Machine-Closed Reconstruction of Social and Collective Sciences from Smithian Fold Theory	15
Abstract	15
Successor headline findings	15
Current status, evidence language and reader map	16
Corpus-wide terminology and evidence key	17
Three reading levels	17
Editorial change control	17
Authorship, access and the public scientific mission	18
Mathematical constitution	18
Empirical constitution and blind order	18
Dependency and ownership boundary	19
Registered primary-source surface	19
Family results	20
Agents, relations, groups and collective identity	20
Observation, action and communication	20
Coordination, cooperation, competition and conflict	21
Norms, rules, institutions and enforcement	21
Trust, reputation and collective knowledge	21
Exchange, allocation and distribution	22
Power, authority, dependency and exclusion	22
Networks, diffusion and collective transition	22
Culture, language and transmission	23
Individual-collective composition and causation	23
History, intervention, replication and ethics	23
Disciplinary ownership and implementation handoffs	24
Complete claim-by-claim derivation record	24
1. Social agent	24
Forward chain and exact carrier	24
Complete enumeration and uniqueness	25
Falsification and deliberately unfavourable controls	25
Post-seal external evidence	26
Scientific meaning	26
Machine certificate	26
2. Social relation	26
Forward chain and exact carrier	27
Complete enumeration and uniqueness	27
Falsification and deliberately unfavourable controls	28
Post-seal external evidence	28
Scientific meaning	28
Machine certificate	29

3. Group identity	29
Forward chain and exact carrier	29
Complete enumeration and uniqueness	29
Falsification and deliberately unfavourable controls	30
Post-seal external evidence	30
Scientific meaning	31
Machine certificate	31
4. Collective state	31
Forward chain and exact carrier	31
Complete enumeration and uniqueness	32
Falsification and deliberately unfavourable controls	32
Post-seal external evidence	32
Scientific meaning	33
Machine certificate	33
5. Organisation identity	33
Forward chain and exact carrier	33
Complete enumeration and uniqueness	34
Falsification and deliberately unfavourable controls	34
Post-seal external evidence	35
Scientific meaning	35
Machine certificate	35
6. Social population	35
Forward chain and exact carrier	35
Complete enumeration and uniqueness	36
Falsification and deliberately unfavourable controls	37
Post-seal external evidence	37
Scientific meaning	37
Machine certificate	37
7. Social observation	37
Forward chain and exact carrier	38
Complete enumeration and uniqueness	38
Falsification and deliberately unfavourable controls	39
Post-seal external evidence	39
Scientific meaning	39
Machine certificate	40
8. Social action	40
Forward chain and exact carrier	40
Complete enumeration and uniqueness	40
Falsification and deliberately unfavourable controls	41
Post-seal external evidence	41
Scientific meaning	42
Machine certificate	42
9. Communication event	42
Forward chain and exact carrier	42
Complete enumeration and uniqueness	43

Falsification and deliberately unfavourable controls	43
Post-seal external evidence	43
Scientific meaning	44
Machine certificate	44
10. Shared social record	44
Forward chain and exact carrier	44
Complete enumeration and uniqueness	45
Falsification and deliberately unfavourable controls	45
Post-seal external evidence	46
Scientific meaning	46
Machine certificate	46
11. Common-knowledge boundary	46
Forward chain and exact carrier	46
Complete enumeration and uniqueness	47
Falsification and deliberately unfavourable controls	48
Post-seal external evidence	48
Scientific meaning	48
Machine certificate	48
12. Social communication noise	49
Forward chain and exact carrier	49
Complete enumeration and uniqueness	49
Falsification and deliberately unfavourable controls	50
Post-seal external evidence	50
Scientific meaning	50
Machine certificate	51
13. Coordination	51
Forward chain and exact carrier	51
Complete enumeration and uniqueness	51
Falsification and deliberately unfavourable controls	52
Post-seal external evidence	52
Scientific meaning	53
Machine certificate	53
14. Cooperation	53
Forward chain and exact carrier	53
Complete enumeration and uniqueness	54
Falsification and deliberately unfavourable controls	54
Post-seal external evidence	55
Scientific meaning	55
Machine certificate	55
15. Competition	55
Forward chain and exact carrier	55
Complete enumeration and uniqueness	56
Falsification and deliberately unfavourable controls	57
Post-seal external evidence	57
Scientific meaning	57

Machine certificate	57
16. Conflict	58
Forward chain and exact carrier	58
Complete enumeration and uniqueness	58
Falsification and deliberately unfavourable controls	59
Post-seal external evidence	59
Scientific meaning	59
Machine certificate	60
17. Bargaining	60
Forward chain and exact carrier	60
Complete enumeration and uniqueness	60
Falsification and deliberately unfavourable controls	61
Post-seal external evidence	61
Scientific meaning	62
Machine certificate	62
18. Collective action	62
Forward chain and exact carrier	62
Complete enumeration and uniqueness	63
Falsification and deliberately unfavourable controls	63
Post-seal external evidence	64
Scientific meaning	64
Machine certificate	64
19. Social norm	64
Forward chain and exact carrier	64
Complete enumeration and uniqueness	65
Falsification and deliberately unfavourable controls	66
Post-seal external evidence	66
Scientific meaning	66
Machine certificate	67
20. Social rule	67
Forward chain and exact carrier	67
Complete enumeration and uniqueness	67
Falsification and deliberately unfavourable controls	68
Post-seal external evidence	68
Scientific meaning	69
Machine certificate	69
21. Institution	69
Forward chain and exact carrier	69
Complete enumeration and uniqueness	70
Falsification and deliberately unfavourable controls	70
Post-seal external evidence	70
Scientific meaning	71
Machine certificate	71
22. Enforcement	71
Forward chain and exact carrier	71

Complete enumeration and uniqueness	72
Falsification and deliberately unfavourable controls	72
Post-seal external evidence	73
Scientific meaning	73
Machine certificate	73
23. Compliance and resistance	73
Forward chain and exact carrier	74
Complete enumeration and uniqueness	74
Falsification and deliberately unfavourable controls	75
Post-seal external evidence	75
Scientific meaning	75
Machine certificate	76
24. Institutional change	76
Forward chain and exact carrier	76
Complete enumeration and uniqueness	76
Falsification and deliberately unfavourable controls	77
Post-seal external evidence	77
Scientific meaning	78
Machine certificate	78
25. Trust relation	78
Forward chain and exact carrier	78
Complete enumeration and uniqueness	79
Falsification and deliberately unfavourable controls	79
Post-seal external evidence	79
Scientific meaning	80
Machine certificate	80
26. Reputation	80
Forward chain and exact carrier	80
Complete enumeration and uniqueness	81
Falsification and deliberately unfavourable controls	81
Post-seal external evidence	82
Scientific meaning	82
Machine certificate	82
27. Testimony and credibility	82
Forward chain and exact carrier	83
Complete enumeration and uniqueness	83
Falsification and deliberately unfavourable controls	84
Post-seal external evidence	84
Scientific meaning	84
Machine certificate	85
28. Collective knowledge	85
Forward chain and exact carrier	85
Complete enumeration and uniqueness	85
Falsification and deliberately unfavourable controls	86
Post-seal external evidence	86

Scientific meaning	87
Machine certificate	87
29. Expertise evidence boundary	87
Forward chain and exact carrier	87
Complete enumeration and uniqueness	88
Falsification and deliberately unfavourable controls	88
Post-seal external evidence	88
Scientific meaning	89
Machine certificate	89
30. False and unsupported social information	89
Forward chain and exact carrier	89
Complete enumeration and uniqueness	90
Falsification and deliberately unfavourable controls	90
Post-seal external evidence	91
Scientific meaning	91
Machine certificate	91
31. Exchange	91
Forward chain and exact carrier	91
Complete enumeration and uniqueness	92
Falsification and deliberately unfavourable controls	93
Post-seal external evidence	93
Scientific meaning	93
Machine certificate	93
32. Social valuation	94
Forward chain and exact carrier	94
Complete enumeration and uniqueness	94
Falsification and deliberately unfavourable controls	95
Post-seal external evidence	95
Scientific meaning	95
Machine certificate	96
33. Resource allocation	96
Forward chain and exact carrier	96
Complete enumeration and uniqueness	96
Falsification and deliberately unfavourable controls	97
Post-seal external evidence	97
Scientific meaning	98
Machine certificate	98
34. Collective production	98
Forward chain and exact carrier	98
Complete enumeration and uniqueness	99
Falsification and deliberately unfavourable controls	99
Post-seal external evidence	99
Scientific meaning	100
Machine certificate	100
35. Distribution and inequality	100

Forward chain and exact carrier	100
Complete enumeration and uniqueness	101
Falsification and deliberately unfavourable controls	101
Post-seal external evidence	102
Scientific meaning	102
Machine certificate	102
36. Social mobility	102
Forward chain and exact carrier	103
Complete enumeration and uniqueness	103
Falsification and deliberately unfavourable controls	104
Post-seal external evidence	104
Scientific meaning	104
Machine certificate	105
37. Power relation	105
Forward chain and exact carrier	105
Complete enumeration and uniqueness	105
Falsification and deliberately unfavourable controls	106
Post-seal external evidence	106
Scientific meaning	107
Machine certificate	107
38. Authority	107
Forward chain and exact carrier	107
Complete enumeration and uniqueness	108
Falsification and deliberately unfavourable controls	108
Post-seal external evidence	108
Scientific meaning	109
Machine certificate	109
39. Social dependency	109
Forward chain and exact carrier	109
Complete enumeration and uniqueness	110
Falsification and deliberately unfavourable controls	110
Post-seal external evidence	111
Scientific meaning	111
Machine certificate	111
40. Exclusion	111
Forward chain and exact carrier	111
Complete enumeration and uniqueness	112
Falsification and deliberately unfavourable controls	113
Post-seal external evidence	113
Scientific meaning	113
Machine certificate	113
41. Stratification	114
Forward chain and exact carrier	114
Complete enumeration and uniqueness	114
Falsification and deliberately unfavourable controls	115

Post-seal external evidence	115
Scientific meaning	115
Machine certificate	116
42. Scientific and institutional gatekeeping	116
Forward chain and exact carrier	116
Complete enumeration and uniqueness	116
Falsification and deliberately unfavourable controls	117
Post-seal external evidence	117
Scientific meaning	118
Machine certificate	118
43. Social network	118
Forward chain and exact carrier	118
Complete enumeration and uniqueness	119
Falsification and deliberately unfavourable controls	119
Post-seal external evidence	120
Scientific meaning	120
Machine certificate	120
44. Social diffusion	120
Forward chain and exact carrier	120
Complete enumeration and uniqueness	121
Falsification and deliberately unfavourable controls	122
Post-seal external evidence	122
Scientific meaning	122
Machine certificate	122
45. Collective synchronization	122
Forward chain and exact carrier	123
Complete enumeration and uniqueness	123
Falsification and deliberately unfavourable controls	124
Post-seal external evidence	124
Scientific meaning	124
Machine certificate	125
46. Collective cascade	125
Forward chain and exact carrier	125
Complete enumeration and uniqueness	125
Falsification and deliberately unfavourable controls	126
Post-seal external evidence	126
Scientific meaning	127
Machine certificate	127
47. Collective transition	127
Forward chain and exact carrier	127
Complete enumeration and uniqueness	128
Falsification and deliberately unfavourable controls	128
Post-seal external evidence	128
Scientific meaning	129
Machine certificate	129

48. Socio-economic cycle	129
Forward chain and exact carrier	129
Complete enumeration and uniqueness	130
Falsification and deliberately unfavourable controls	130
Post-seal external evidence	131
Scientific meaning	131
Machine certificate	131
49. Culture	131
Forward chain and exact carrier	132
Complete enumeration and uniqueness	132
Falsification and deliberately unfavourable controls	133
Post-seal external evidence	133
Scientific meaning	133
Machine certificate	134
50. Social symbol and meaning	134
Forward chain and exact carrier	134
Complete enumeration and uniqueness	134
Falsification and deliberately unfavourable controls	135
Post-seal external evidence	135
Scientific meaning	136
Machine certificate	136
51. Language community	136
Forward chain and exact carrier	136
Complete enumeration and uniqueness	137
Falsification and deliberately unfavourable controls	137
Post-seal external evidence	137
Scientific meaning	138
Machine certificate	138
52. Education and knowledge transmission	138
Forward chain and exact carrier	138
Complete enumeration and uniqueness	139
Falsification and deliberately unfavourable controls	139
Post-seal external evidence	140
Scientific meaning	140
Machine certificate	140
53. Media system	140
Forward chain and exact carrier	141
Complete enumeration and uniqueness	141
Falsification and deliberately unfavourable controls	142
Post-seal external evidence	142
Scientific meaning	142
Machine certificate	143
54. Collective memory	143
Forward chain and exact carrier	143
Complete enumeration and uniqueness	143

Falsification and deliberately unfavourable controls	144
Post-seal external evidence	144
Scientific meaning	145
Machine certificate	145
55. Individual-to-collective composition	145
Forward chain and exact carrier	145
Complete enumeration and uniqueness	146
Falsification and deliberately unfavourable controls	146
Post-seal external evidence	147
Scientific meaning	147
Machine certificate	147
56. Collective-to-individual constraint	147
Forward chain and exact carrier	147
Complete enumeration and uniqueness	148
Falsification and deliberately unfavourable controls	149
Post-seal external evidence	149
Scientific meaning	149
Machine certificate	149
57. Collective emergence	150
Forward chain and exact carrier	150
Complete enumeration and uniqueness	150
Falsification and deliberately unfavourable controls	151
Post-seal external evidence	151
Scientific meaning	151
Machine certificate	152
58. Human and artificial multi-agent collective	152
Forward chain and exact carrier	152
Complete enumeration and uniqueness	152
Falsification and deliberately unfavourable controls	153
Post-seal external evidence	153
Scientific meaning	154
Machine certificate	154
59. Collective decision	154
Forward chain and exact carrier	154
Complete enumeration and uniqueness	155
Falsification and deliberately unfavourable controls	155
Post-seal external evidence	156
Scientific meaning	156
Machine certificate	156
60. Aggregation and ecological inference boundary	156
Forward chain and exact carrier	156
Complete enumeration and uniqueness	157
Falsification and deliberately unfavourable controls	158
Post-seal external evidence	158
Scientific meaning	158

Machine certificate	158
61. Historical social event	158
Forward chain and exact carrier	159
Complete enumeration and uniqueness	159
Falsification and deliberately unfavourable controls	160
Post-seal external evidence	160
Scientific meaning	160
Machine certificate	161
62. Historical contingency	161
Forward chain and exact carrier	161
Complete enumeration and uniqueness	161
Falsification and deliberately unfavourable controls	162
Post-seal external evidence	162
Scientific meaning	163
Machine certificate	163
63. Social causal inference	163
Forward chain and exact carrier	163
Complete enumeration and uniqueness	164
Falsification and deliberately unfavourable controls	164
Post-seal external evidence	164
Scientific meaning	165
Machine certificate	165
64. Social intervention	165
Forward chain and exact carrier	165
Complete enumeration and uniqueness	166
Falsification and deliberately unfavourable controls	166
Post-seal external evidence	167
Scientific meaning	167
Machine certificate	167
65. Cross-context replication	167
Forward chain and exact carrier	168
Complete enumeration and uniqueness	168
Falsification and deliberately unfavourable controls	169
Post-seal external evidence	169
Scientific meaning	169
Machine certificate	170
66. Ethical and value boundary	170
Forward chain and exact carrier	170
Complete enumeration and uniqueness	170
Falsification and deliberately unfavourable controls	171
Post-seal external evidence	171
Scientific meaning	172
Machine certificate	172
67. Mathematics and graph handoff	172
Forward chain and exact carrier	172

Complete enumeration and uniqueness	173
Falsification and deliberately unfavourable controls	173
Post-seal external evidence	173
Scientific meaning	174
Machine certificate	174
68. Computation and distributed-system handoff	174
Forward chain and exact carrier	174
Complete enumeration and uniqueness	175
Falsification and deliberately unfavourable controls	175
Post-seal external evidence	176
Scientific meaning	176
Machine certificate	176
69. Biology, consciousness and individual cognition handoff	176
Forward chain and exact carrier	177
Complete enumeration and uniqueness	177
Falsification and deliberately unfavourable controls	178
Post-seal external evidence	178
Scientific meaning	178
Machine certificate	179
70. Medicine and public-health handoff	179
Forward chain and exact carrier	179
Complete enumeration and uniqueness	179
Falsification and deliberately unfavourable controls	180
Post-seal external evidence	180
Scientific meaning	181
Machine certificate	181
71. Earth and environmental handoff	181
Forward chain and exact carrier	181
Complete enumeration and uniqueness	182
Falsification and deliberately unfavourable controls	182
Post-seal external evidence	183
Scientific meaning	183
Machine certificate	183
72. Engineering and policy implementation handoff	183
Forward chain and exact carrier	183
Complete enumeration and uniqueness	184
Falsification and deliberately unfavourable controls	185
Post-seal external evidence	185
Scientific meaning	185
Machine certificate	186
Historical reconciliation	186
What this foundation does and does not claim	186
Full-field continuation roadmap	186
Reproduction and present validation state	186
Rights, participation and scientific admission	187

Machine identities	187
Complete current-census successor supplement	187
Exact preferential-flow inequality	187
Consensus/polarization Fold lock	188
Dissipative social period-two cycle	188
Complete post-seal exact Social comparison	189
Preceding-version abstract preserved for chronology	190
Registered source surface	190
Limitations and open frontier	190
Data and code availability	190
References and source registry	191
Successor machine-identity appendix	191
Lean 4 independent verification update	191
Verified result	191
Native theorem proof and artifact verification are different layers	192
Fail-closed custody event	192
What the result supports	192
Successor conclusion	192

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<i>Publication control. This is a new versioned successor. It does not rewrite the preceding paper, its DOI record, historical receipts or evidence. No push, upload, DOI action, release or publication is authorised until Maria Smith explicitly confirms the exact final files and hashes.</i>
<i>Lean verification integration. The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.</i>

Abstract

This successor presents the complete current `social_collective_systems`, `social_collective` paper surface: 76 live model-admitted claims, 19,456 generated candidates, 76 unique survivors and 304 passed controls. It integrates 4 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned PASS for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.

3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 76 current claim(s) in its declared branch ownership boundary; 4 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002 - Exact preferential-flow inequality; SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002 - Consensus/polarization Fold lock; SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002 - Dissipative social period-two cycle; SFT-SOCIAL-VALIDATION-EXACT-COMPLETE-FAMILY-002 - Complete post-seal exact Social comparison.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	76 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.1.0 is a final local publication candidate awaiting Maria Smith's approval. It is not deposited or published.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Authorship, access and the public scientific mission

Maria Smith produced this work outside credentialed institutional access. That fact is not offered as personal exceptionalism. It is an indictment of all the minds and contributions society loses when financial gatekeeping, credentials, paywalls, prestige, institutional permission and access to funded infrastructure are dressed as rigour. The appropriate response is not to ask whether an excluded author was somehow uniquely permitted to think. It is to ask how many minds, observations and corrections were never heard because the gate was mistaken for the method.

Ernos Labs is an open-source science movement and a standards designation. The repository, derivations, source failures, controls, papers and machine receipts are public so that any person can inspect, challenge, reproduce or extend them. Transparent access is not a relaxation of rigour: it makes rigour inspectable. Paywalled claims, opaque oracles, credential votes and funding-driven restrictions cannot substitute for a complete derivation, a sealed prediction, an unfavourable control or an independently reproducible result.

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Open criticism requires no credential. Scientific admission requires the common evidence protocol. Contact: **Maria.Smith.Sftoe@gmail.com**. Public submissions and review: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

Mathematical constitution

Every law traces to **There Is No Nothing** and to already admitted Fold dependencies. No additional axiom is introduced. No free or fitted parameter can select a survivor. Proof-state arithmetic is exact and finite. Conventional numerical zero is not a proof carrier; the glyph 0 may be retained as a typed marker for absence, non-response or a source field, but it does not import ordinary zero semantics. Negative, irrational and imaginary quantities may appear as held source labels where an external dataset uses them, but never as proof scalars. Completed infinities and ungenerated continua are excluded.

For each claim the generator builds eight binary dimensions: carrier, boundary, relation, record, evidence, provenance, generality and extension. Their literal Cartesian product contains $2^8 = 256$ candidates. One choice on each dimension alone preserves its required distinction. The program decides all 256 candidates; one survives and 255 are eliminated. Across 72 claims this is exactly 18,432 decided forms. A separate implementation rebuilds the product and survivor from the declared axes. Positive-finite induction proves that adding one lawful agent, relation, context, period or record preserves all earlier identities, adverse rows and boundaries while appending its trace.

This closure is exact but bounded. It establishes the foundational forms necessary to make Social claims reconstructible and falsifiable. It does not prohibit new societies, datasets, institutions, forms of exclusion, historical sources or lawful discoveries. The branch is therefore current-evidence closed and extension-open, never permanently locked.

Empirical constitution and blind order

The complete inventory, candidate grammar, predicted structural labels and falsification conditions were frozen before any external source identity or outcome was opened. Primary custodians were selected afterward for their custody of relevant observation, survey, demographic, conflict, institutional, health, education, publication and historical records-not because their institutional status could prove a law. A capability-closed prediction process cannot read a file, network, clock, environment variable, process or target. Only after its derivation seal exists may the external evaluator open the registered target.

Every comparison retains successful, adverse, absent, censored, missing, disputed, unresolved, access-denied and transport-failed rows. An inaccessible archive is not evidence of no event. A source's classification is not automatically truth. A survey response is not a private mental state. An institutional index is not the institution itself. A model, estimate or forecast remains a model, estimate or forecast. These distinctions are part of the empirical result, not administrative bookkeeping.

The immutable engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a. The

verification-authority seal is

sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. A mismatch halts. Those identities secure a shared admission route; they never replace the readable derivations.

Dependency and ownership boundary

Mathematics owns exact graph, order, probability and optimisation structures. Computation owns abstract communication, causality, consensus and distributed knowledge. Biology owns organism-level processes. Consciousness owns first-person experience and qualia. Medicine owns clinical state. Earth Science owns physical hazards and environments. Social and Collective Sciences owns measured relations, group membership, institutions, distribution, collective action, historical context, public access and social consequence. Engineering Translation may implement and test applications but cannot select or rewrite the laws. This ownership map prevents an abstract edge from becoming a social tie, a simulated agent from becoming a human observation, or a product metric from becoming scientific admission.

Registered primary-source surface

- **UNSD-DEMOGRAPHIC-001 - United Nations Statistics Division.** Demographic and Social Statistics. Registered families: agents_groups_identity, exchange_allocation_distribution, power_authority_exclusion. Locator: https://unstats.un.org/unsd/demographic-social/products/dyb/dyb_2024/. Its authority here is source custody and documented provenance, not prestige or consensus.
- **WORLD-BANK-WDI-001 - World Bank Data Group.** World Development Indicators. Registered families: exchange_allocation_distribution, power_authority_exclusion, multi_level_causation. Locator: <https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **ILOSTAT-001 - International Labour Organization Department of Statistics.** ILOSTAT data. Registered families: exchange_allocation_distribution, power_authority_exclusion, agents_groups_identity. Locator: <https://ilostat.ilo.org/data/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UNESCO-UIS-001 - UNESCO Institute for Statistics.** UIS Data Browser. Registered families: culture_language_transmission, exchange_allocation_distribution, power_authority_exclusion. Locator: <https://databrowser.uis.unesco.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UCDP-DOWNLOADS-001 - Uppsala Conflict Data Program.** UCDP Dataset Download Center. Registered families: coordination_conflict, history_intervention_ethics, power_authority_exclusion. Locator: <https://ucdp.uu.se/downloads/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **VDEM-DATA-001 - Varieties of Democracy Institute.** V-Dem Dataset Archive. Registered families: norm_institution_enforcement, power_authority_exclusion, history_intervention_ethics. Locator: <https://v-dem.net/data/dataset-archive/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **IPUMS-INTERNATIONAL-001 - IPUMS / University of Minnesota.** IPUMS International. Registered families: agents_groups_identity, exchange_allocation_distribution, culture_language_transmission, multi_level_causation. Locator: <https://www.ipums.org/projects/ipums-international>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **GSS-NORC-001 - NORC at the University of Chicago.** General Social Survey. Registered families: trust_reputation_knowledge, culture_language_transmission, observation_action_communication. Locator: <https://gss.norc.umd.edu/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **OECD-DATA-EXPLORER-001 - Organisation for Economic Co-operation and Development.** OECD Data Explorer. Registered families: exchange_allocation_distribution, norm_institution_enforcement, multi_level_causation. Locator: <https://data-explorer.oecd.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **UNHCR-REFUGEE-DATA-001 - United Nations High Commissioner for Refugees.** Refugee Data Finder. Registered families: agents_groups_identity, power_authority_exclusion, history_intervention_ethics. Locator: <https://www.unhcr.org/refugee-statistics/>. Its authority here is source custody and documented provenance, not prestige or

consensus.

- **UNODC-DATA-001 - United Nations Office on Drugs and Crime.** UNODC Data Portal. Registered families: norm_institution_enforcement, coordination_conflict, history_intervention_ethics. Locator: <https://dataunodc.un.org/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **WHO-GHO-001 - World Health Organization.** Global Health Observatory. Registered families: multi_level_causation, ownership_handoffs, exchange_allocation_distribution. Locator: <https://www.who.int/data/gho>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **ICPSR-DATA-001 - Inter-university Consortium for Political and Social Research.** ICPSR Data Archive. Registered families: observation_action_communication, trust_reputation_knowledge, history_intervention_ethics. Locator: <https://www.icpsr.umich.edu/web/pages/>. Its authority here is source custody and documented provenance, not prestige or consensus.
- **CROSSREF-RECORDS-001 - Crossref.** Crossref metadata and records. Registered families: trust_reputation_knowledge, culture_language_transmission, ownership_handoffs. Locator: <https://www.crossref.org/documentation/retrieve-metadata/rest-api/>. Its authority here is source custody and documented provenance, not prestige or consensus.

Family results

Agents, relations, groups and collective identity

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social agent** (SFT-SOCIAL-AGENT-001): A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit.
- **Social relation** (SFT-SOCIAL-RELATION-001): A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context.
- **Group identity** (SFT-SOCIAL-GROUP-001): A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person.
- **Collective state** (SFT-SOCIAL-COLLECTIVE-001): A collective state composes agent states and relations while retaining internal distribution and minority states.
- **Organization identity** (SFT-SOCIAL-ORGANIZATION-001): An organization is a versioned arrangement of members, roles, rules, resources and processes.
- **Social population** (SFT-SOCIAL-POPULATION-001): A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained.

Observation, action and communication

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social observation** (SFT-SOCIAL-OBSERVATION-001): Social observation retains observer, method, consent, setting, reactivity and unavailable records.
- **Social action** (SFT-SOCIAL-ACTION-001): Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence.
- **Communication event** (SFT-SOCIAL-MESSAGE-001): A communication event retains sender, content, channel, recipient opportunity, reception and loss.
- **Shared social record** (SFT-SOCIAL-SHARED-RECORD-001): A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained.
- **Common-knowledge boundary** (SFT-SOCIAL-COMMON-KNOWLEDGE-001): Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain.

- **Social communication noise** (SFT-SOCIAL-COMMUNICATION-NOISE-001): Communication noise retains technical loss, ambiguity, reinterpretation and audience differences.

Coordination, cooperation, competition and conflict

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Coordination** (SFT-SOCIAL-COORDINATION-001): Coordination is compatible action around a declared target with dependencies, timing and failures retained.
- **Cooperation** (SFT-SOCIAL-COOPERATION-001): Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants.
- **Competition** (SFT-SOCIAL-COMPETITION-001): Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained.
- **Conflict** (SFT-SOCIAL-CONFLICT-001): Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time.
- **Bargaining** (SFT-SOCIAL-BARGAINING-001): Bargaining retains offers, outside options, information, power, time, breakdown and agreement.
- **Collective action** (SFT-SOCIAL-COLLECTIVE-ACTION-001): Collective action retains participation, abstention, contribution, threshold, benefit, failure and context.

Norms, rules, institutions and enforcement

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social norm** (SFT-SOCIAL-NORM-001): A norm is a context-bound expectation evidenced by behavior and responses to conformity and violation.
- **Social rule** (SFT-SOCIAL-RULE-001): A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation.
- **Institution** (SFT-SOCIAL-INSTITUTION-001): An institution is a historically repeated governed process, not merely a building, name or consensus.
- **Enforcement** (SFT-SOCIAL-ENFORCEMENT-001): Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement.
- **Compliance and resistance** (SFT-SOCIAL-COMPLIANCE-001): Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence.
- **Institutional change** (SFT-SOCIAL-INSTITUTIONAL-CHANGE-001): Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts.

Trust, reputation and collective knowledge

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Trust relation** (SFT-SOCIAL-TRUST-001): Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained.
- **Reputation** (SFT-SOCIAL-REPUTATION-001): Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible.
- **Testimony and credibility** (SFT-SOCIAL-TESTIMONY-001): Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution.
- **Collective knowledge** (SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001): Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained.

- **Expertise evidence boundary** (SFT-SOCIAL-EXPERTISE-001): Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token.
- **False and unsupported social information** (SFT-SOCIAL-MISINFORMATION-001): False, unsupported, disputed and corrected claims remain distinct and source-dated.

Exchange, allocation and distribution

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Exchange** (SFT-SOCIAL-EXCHANGE-001): Exchange retains parties, items, rights, terms, timing, costs and failed transfers.
- **Social valuation** (SFT-SOCIAL-VALUE-001): Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration.
- **Resource allocation** (SFT-SOCIAL-ALLOCATION-001): Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses.
- **Collective production** (SFT-SOCIAL-PRODUCTION-001): Production retains labour, material, knowledge, ownership, transformation, output and loss.
- **Distribution and inequality** (SFT-SOCIAL-DISTRIBUTION-001): Distribution retains population, total stock, shares, access, time and missing members.
- **Social mobility** (SFT-SOCIAL-MOBILITY-001): Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure.

Power, authority, dependency and exclusion

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Power relation** (SFT-SOCIAL-POWER-001): Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance.
- **Authority** (SFT-SOCIAL-AUTHORITY-001): Authority is scoped decision capacity with source, procedure, recognition and contestation retained.
- **Social dependency** (SFT-SOCIAL-DEPENDENCY-001): Dependency retains the needed resource, controller, alternatives, costs and failures.
- **Exclusion** (SFT-SOCIAL-EXCLUSION-001): Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases.
- **Stratification** (SFT-SOCIAL-STRATIFICATION-001): Stratification is persistent ordered access across a declared population and resource set with mobility retained.
- **Scientific and institutional gatekeeping** (SFT-SOCIAL-GATEKEEPING-001): Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion.

Networks, diffusion and collective transition

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Social network** (SFT-SOCIAL-SOCIAL-NETWORK-001): A social network retains node identity, link meaning, direction, time, sampling and missing links.
- **Social diffusion** (SFT-SOCIAL-DIFFUSION-001): Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs.
- **Collective synchronization** (SFT-SOCIAL-SYNCHRONIZATION-001): Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained.

- **Collective cascade** (SFT-SOCIAL-CASCADE-001): A cascade retains ordered local transitions, paths, nontransitions and external causes.
- **Collective transition** (SFT-SOCIAL-COLLECTIVE-TRANSITION-001): Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained.
- **Socio-economic cycle** (SFT-SOCIAL-SOCIAL-CYCLE-001): A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change.

Culture, language and transmission

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Culture** (SFT-SOCIAL-CULTURE-001): Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group.
- **Social symbol and meaning** (SFT-SOCIAL-SYMBOL-001): A symbol's social meaning is user-, context- and history-bound with alternatives retained.
- **Language community** (SFT-SOCIAL-LANGUAGE-COMMUNITY-001): A language community retains speakers, varieties, contexts, power, change and corpus limits.
- **Education and knowledge transmission** (SFT-SOCIAL-EDUCATION-001): Education evidence retains access, instruction, participation, assessment, attrition and context.
- **Media system** (SFT-SOCIAL-MEDIA-001): A media system retains ownership, production, selection, distribution, reach, response and missing exposure.
- **Collective memory** (SFT-SOCIAL-COLLECTIVE-MEMORY-001): Collective memory is a contested transmission of records, narratives and practices with omissions retained.

Individual-collective composition and causation

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Individual-to-collective composition** (SFT-SOCIAL-MICRO-MACRO-001): Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions.
- **Collective-to-individual constraint** (SFT-SOCIAL-MACRO-MICRO-001): Collective structures affect individuals through observable rules, resources, exposure and expectations.
- **Collective emergence** (SFT-SOCIAL-EMERGENCE-001): An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained.
- **Human and artificial multi-agent collective** (SFT-SOCIAL-MULTI-AGENT-001): A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight.
- **Collective decision** (SFT-SOCIAL-COLLECTIVE-DECISION-001): Collective decision retains eligibility, options, information, rule, participation, dissent and implementation.
- **Aggregation and ecological inference boundary** (SFT-SOCIAL-AGGREGATION-LOSS-001): Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals.

History, intervention, replication and ethics

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Historical social event** (SFT-SOCIAL-HISTORICAL-EVENT-001): A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts.
- **Historical contingency** (SFT-SOCIAL-CONTINGENCY-001): Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals.

- **Social causal inference** (SFT-SOCIAL-CAUSAL-INFERENCE-001): Causal inference retains assignment, comparison, confounding, spillover, attrition and target population.
- **Social intervention** (SFT-SOCIAL-INTERVENTION-001): Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes.
- **Cross-context replication** (SFT-SOCIAL-REPLICATION-001): Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits.
- **Ethical and value boundary** (SFT-SOCIAL-ETHICS-BOUNDARY-001): Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law.

Disciplinary ownership and implementation handoffs

This family contains 6 separately admitted obligations. Its closure preserves the level of analysis, the context and the complete record; it does not universalise one population or historical period.

- **Mathematics and graph handoff** (SFT-SOCIAL-MATH-HANDOFF-001): Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity.
- **Computation and distributed-system handoff** (SFT-SOCIAL-COMPUTATION-HANDOFF-001): Computation may simulate a declared social process but cannot make simulated behavior an observation.
- **Biology, consciousness and individual cognition handoff** (SFT-SOCIAL-BIO-CONSC-HANDOFF-001): Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations.
- **Medicine and public-health handoff** (SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001): Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both.
- **Earth and environmental handoff** (SFT-SOCIAL-EARTH-HANDOFF-001): Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact.
- **Engineering and policy implementation handoff** (SFT-SOCIAL-ENGINEERING-HANDOFF-001): Engineering may test implementations against social laws but applications cannot select or rewrite those laws.

Complete claim-by-claim derivation record

The following seventy-two sections are the human-readable chain. The machine packages preserve all 256 candidate rows, 256 decisions, hostile controls, source manifest, independent output, empirical receipt and engine receipt for each claim.

1. Social agent

Claim: SFT-SOCIAL-AGENT-001

Family: agents_groups_identity

Exact statement: A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit.

Forward chain and exact carrier

The generated carrier is bounded-acting-observed-agent-carrier. Its required relation is state-action-consequence-relation. The reconstruction must retain identity-time-capability-context-and-alternatives-held. Its declared empirical boundary is agent-identification-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,

SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: bounded-acting-observed-agent-carrier.
- **boundary** - which population, place, period and method bound the statement: agent-identification-bounded.
- **relation** - which generated interaction or comparison connects the carriers: state-action-consequence-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: identity-time-capability-context-and-alternatives-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
bounded-acting-observed-agent-carrier__agent-identification-bounded__state-action-consequence-relation__identity-time-capability-context-and-alternatives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a person, role, organisation or artificial agent is silently substituted for another.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social agent**, the consequence is practical and exact: A social agent is a bounded acting and observed carrier whose identity, capability, context and alternatives remain explicit. A scientific claim therefore has to expose the carrier bounded-acting-observed-agent-carrier, the relation state-action-consequence-relation, the record identity-time-capability-context-and-alternatives-held, and the boundary agent-identification-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:9aa0e286281d1e575e03574ed87996d5d10b51642fb320803a30af7d72f6f241; independent
certificate: sha256:7f0c74c6bdca1b38f8101cb8f52eeb417ad5401cb1039e661671e0313b337136; engine
receipt: sha256:f2f609e054ce3520a7f646913f90e57d7dc53aaaf77ba0556570d0ff27c17ebd. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

2. Social relation

Claim: SFT-SOCIAL-RELATION-001

Family: agents_groups_identity

Exact statement: A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context.

Forward chain and exact carrier

The generated carrier is `ordered-agent-pair-context-carrier`. Its required relation is `typed-interaction-relation`. The reconstruction must retain `participants-direction-content-time-and-context-held`. Its declared empirical boundary is `relation-observation-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-AGENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `ordered-agent-pair-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `relation-observation-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `typed-interaction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-direction-content-time-and-context-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
ordered-agent-pair-context-carrier__relation-observation-bounded__typed-interaction-r
elation__participants-direction-content-time-and-context-held__evidence-and-normative
-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-co
ntexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes

one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if co-presence, correlation or one-sided reporting proves a reciprocal relation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social relation**, the consequence is practical and exact: A social relation is a typed, directed or reciprocal interaction between identified agents in a declared context. A scientific claim therefore has to expose the carrier `ordered-agent-pair-context-carrier`, the relation `typed-interaction-relation`, the record `participants-direction-content-time-and-context-held`, and the boundary `relation-observation-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 9aeb0a35ead80c526e186c70bd144294fe43470a110ec9d38e00564c25c1a065; independent certificate: sha256: 9c461b66d85b8d19d19fa6877ced25088ab93b9e57772bfeb1f69f9c70571d12; engine receipt: sha256: 146c15af99c64010363b325bf1d0df656f6680492ca32f2fce517b83adf8b7cf. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

3. Group identity

Claim: SFT-SOCIAL-GROUP-001

Family: agents_groups_identity

Exact statement: A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person.

Forward chain and exact carrier

The generated carrier is members-boundary-purpose-carrier. Its required relation is membership-and-coordination-relation. The reconstruction must retain members-entry-exit-purpose-time-and-disagreement-held. Its declared empirical boundary is group-definition-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-RELATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: members-boundary-purpose-carrier.
- **boundary** - which population, place, period and method bound the statement: group-definition-bounded.
- **relation** - which generated interaction or comparison connects the carriers: membership-and-coordination-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: members-entry-exit-purpose-time-and-disagreement-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
members-boundary-purpose-carrier__group-definition-bounded__membership-and-coordination-relation__members-entry-exit-purpose-time-and-disagreement-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a label or sample is treated as a coherent group without membership evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **group identity**, the consequence is practical and exact: A group retains membership, boundary, purpose, entry, exit and internal disagreement rather than becoming an indivisible person. A scientific claim therefore has to expose the carrier members-boundary-purpose-carrier, the relation membership-and-coordination-relation, the record members-entry-exit-purpose-time-and-disagreement-held, and the boundary group-definition-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:3d94b6fca6fa1a8a0da49d56d63ff1374029f33e773144ad683bd15369d227ac; independent certificate: sha256:e916c9cca85a24f571f5df78b18f2274cd5a12ef6091e1e292c0662c2d203266; engine receipt: sha256:ea38d831ac38a601d2aaedcfa5153d810f8ca7d31b6227d420da9b8c6956f387. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

4. Collective state

Claim: SFT-SOCIAL-COLLECTIVE-001

Family: agents_groups_identity

Exact statement: A collective state composes agent states and relations while retaining internal distribution and minority states.

Forward chain and exact carrier

The generated carrier is agents-relations-shared-state-carrier. Its required relation is joint-state-composition-relation. The reconstruction must retain members-links-distribution-time-and-minority-states-held. Its declared empirical boundary is collective-observation-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GROUP-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-relations-shared-state-carrier`.
- **boundary** - which population, place, period and method bound the statement: `collective-observation-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `joint-state-composition-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `members-links-distribution-time-and-minority-states-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-relations-shared-state-carrier__collective-observation-bounded__joint-state-composition-relation__members-links-distribution-time-and-minority-states-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an aggregate erases all constituent variation or becomes an individual mind.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `UNSD-DEMOGRAPHIC-001` - **transport** captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective state**, the consequence is practical and exact: A collective state composes agent states and relations while retaining internal distribution and minority states. A scientific claim therefore has to expose the carrier agents-relations-shared-state-carrier, the relation joint-state-composition-relation, the record members-links-distribution-time-and-minority-states-held, and the boundary collective-observation-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1e213b8e450f2ca5fd440a09b4b06614593d6db3840afb5c2fd3f8d232c025f9; independent certificate: sha256:5759ffd91cce60f60a36ee94cd86abec93509cccca90880d27c1f398ff9a8063; engine receipt: sha256:f0866c38e5fc8d9c9dc0190ddb5a8994de2b679bddf4406a43141629c5ee9cda. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

5. Organisation identity

Claim: SFT-SOCIAL-ORGANIZATION-001

Family: agents_groups_identity

Exact statement: An organization is a versioned arrangement of members, roles, rules, resources and processes.

Forward chain and exact carrier

The generated carrier is roles-members-rules-resources-carrier. Its required relation is role-and-process-organization. The reconstruction must retain membership-authority-rules-resources-time-and-version-held. Its declared empirical boundary is organization-boundary-declared. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-001. Each dependency was model-admitted before this claim. The present claim adds no

axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `roles-members-rules-resources-carrier`.
- **boundary** - which population, place, period and method bound the statement: `organization-boundary-declared`.
- **relation** - which generated interaction or comparison connects the carriers: `role-and-process-organization`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `membership-authority-rules-resources-time-and-version-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
roles-members-rules-resources-carrier__organization-boundary-declared__role-and-process-organization__membership-authority-rules-resources-time-and-version-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a legal name alone establishes stable organisation or unchanged purpose.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 1. Unresolved transports retained: 2.
The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or
prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation:
false.

- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and
preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **organisation identity**, the consequence is practical and exact: An organisation is a versioned arrangement of members, roles, rules, resources and processes. A scientific claim therefore has to expose the carrier
roles-members-rules-resources-carrier, the relation role-and-process-organization, the record
membership-authority-rules-resources-time-and-version-held, and the boundary
organization-boundary-declared together. If any one is hidden, the result may remain a description, interpretation,
model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:b51f090e2e0186c60ae25ee38e3d8a67f989a0feab7d1b8623fff59970ee93d4; independent
certificate: sha256:a360f569326af227f28930803909d1950580ff649c67ab4d521f929a620dac65; engine
receipt: sha256:422c2814a0efc03176f88be51639b782eb4bf45afb9cd4dbf9bdd83660547e9e. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

6. Social population

Claim: SFT-SOCIAL-POPULATION-001

Family: agents_groups_identity

Exact statement: A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained.

Forward chain and exact carrier

The generated carrier is selection-bounded-agent-set-carrier. Its required relation is
membership-distribution-and-weighting-relation. The reconstruction must retain
frame-selection-time-weights-censoring-and-missingness-held. Its declared empirical boundary is
population-sampling-bounded. These coordinates are not labels added after a result: they were part of the sealed
obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
 SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
 SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
 SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,
 SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,
 SFT-SOCIAL-ORGANIZATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: selection-bounded-agent-set-carrier.
- **boundary** - which population, place, period and method bound the statement: population-sampling-bounded.
- **relation** - which generated interaction or comparison connects the carriers:
membership-distribution-and-weighting-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
frame-selection-time-weights-censoring-and-missingness-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

selection-bounded-agent-set-carrier__population-sampling-bounded__membership-distribution-and-weighting-relation__frame-selection-time-weights-censoring-and-missingness-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject universal inference from favourable respondents or an undefined denominator.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social population**, the consequence is practical and exact: A social population is a sampling-frame-bound agent set with selection, weighting, censoring and missingness retained. A scientific claim therefore has to expose the carrier `selection-bounded-agent-set-carrier`, the relation `membership-distribution-and-weighting-relation`, the record `frame-selection-time-weights-censoring-and-missingness-held`, and the boundary `population-sampling-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:9bbb9038c173f4f3295f99cf90335d41bc8b21f4f7c359a2eb68300046d279c8`; independent certificate: `sha256:c9e4cf25500edf37bde6e3b927a773ebc772182edbb4b05f978467cb7e32e401`; engine receipt: `sha256:176f9f0a8f0c176a6beca49d36d60c4310ba772a229681fc2eb7a7c3f6187ece`. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

7. Social observation

Claim: SFT-SOCIAL-OBSERVATION-001

Family: `observation_action_communication`

Exact statement: Social observation retains observer, method, consent, setting, reactivity and unavailable records.

Forward chain and exact carrier

The generated carrier is `observer-agent-setting-record-carrier`. Its required relation is `observation-and-reactivity-relation`. The reconstruction must retain `observer-method-consent-setting-time-and-reactivity-held`. Its declared empirical boundary is `observation-method-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-POPULATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `observer-agent-setting-record-carrier`.
- **boundary** - which population, place, period and method bound the statement: `observation-method-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `observation-and-reactivity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `observer-method-consent-setting-time-and-reactivity-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`observer-agent-setting-record-carrier__observation-method-bounded__observation-and-reactivity-relation__observer-method-consent-setting-time-and-reactivity-held__evidence`


```
-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-
retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if observation is treated as intervention-free or private inference as direct report.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social observation**, the consequence is practical and exact: Social observation retains observer, method, consent, setting, reactivity and unavailable records. A scientific claim therefore has to expose the carrier `observer-agent-setting-record-carrier`, the relation `observation-and-reactivity-relation`, the record `observer-method-consent-setting-time-and-reactivity-held`, and the boundary `observation-method-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a

replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:0b0f6e6e5248c46d820a7322e33a4fee588ed491c70574c283f3b627ffa4f3de; independent certificate: sha256:9a174fd7e40a98391357df6582c5ecf55eeb5ecaa9ba8672bfec06901b367377; engine receipt: sha256:9e5f48e91a6767b794854f5c461addb7d9016687a4721cc0970c23fa475f47a6. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

8. Social action

Claim: SFT-SOCIAL-ACTION-001

Family: observation_action_communication

Exact statement: Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence.

Forward chain and exact carrier

The generated carrier is agent-choice-context-consequence-carrier. Its required relation is capability-to-action-relation. The reconstruction must retain alternatives-constraints-intent-act-outcome-and-time-held. Its declared empirical boundary is action-attribution-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-OBSERVATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agent-choice-context-consequence-carrier.
- **boundary** - which population, place, period and method bound the statement: action-attribution-bounded.
- **relation** - which generated interaction or comparison connects the carriers: capability-to-action-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: alternatives-constraints-intent-act-outcome-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
agent-choice-context-consequence-carrier__action-attribution-bounded__capability-to-a
ction-relation__alternatives-constraints-intent-act-outcome-and-time-held__evidence-a
nd-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-re
tains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if outcome alone proves intent or if constrained conduct is treated as free choice.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social action**, the consequence is practical and exact: Action attribution retains capability, alternatives, constraints, intent evidence, act and consequence. A scientific claim therefore has to expose the carrier agent-choice-context-consequence-carrier, the relation capability-to-action-relation, the record alternatives-constraints-intent-act-outcome-and-time-held, and the boundary action-attribution-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 9d8810a1edd01cb9a641eee2f25490238fd37c8c63e47fa7918e62b046349410; independent certificate: sha256: 110df5fbd71c4b1284afc207d5fa3d8e63ede169d8f091573229f72401285d2d; engine receipt: sha256: 99446d01ce954f8802de92a9839ef59588352022b9b9b8665cc6a5ecd70d0c82. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

9. Communication event

Claim: SFT-SOCIAL-MESSAGE-001

Family: observation_action_communication

Exact statement: A communication event retains sender, content, channel, recipient opportunity, reception and loss.

Forward chain and exact carrier

The generated carrier is sender-message-channel-recipient-carrier. Its required relation is encoding-transmission-reception-relation. The reconstruction must retain source-content-channel-time-reach-and-loss-held. Its declared empirical boundary is communication-event-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-ACTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `sender-message-channel-recipient-carrier`.
- **boundary** - which population, place, period and method bound the statement: `communication-event-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `encoding-transmission-reception-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-content-channel-time-reach-and-loss-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
sender-message-channel-recipient-carrier__communication-event-bounded__encoding-transmission-reception-relation__source-content-channel-time-reach-and-loss-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if sending proves receipt, receipt proves belief, or exposure proves action.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **communication event**, the consequence is practical and exact: A communication event retains sender, content, channel, recipient opportunity, reception and loss. A scientific claim therefore has to expose the carrier sender-message-channel-recipient-carrier, the relation encoding-transmission-reception-relation, the record source-content-channel-time-reach-and-loss-held, and the boundary communication-event-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1116cbb366ffcb0be538cf43a2bb7d616432e01edfed1dad0d81508a02afdc44; independent certificate: sha256:5cedd8e07c5660ebef3b556d9095066a77608675951c6256b68fb1d48324ef8e; engine receipt: sha256:f3674a401671a9f62232868be604578e8d23a1b629982ac17a0a58250f24450e. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

10. Shared social record

Claim: SFT-SOCIAL-SHARED-RECORD-001

Family: observation_action_communication

Exact statement: A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained.

Forward chain and exact carrier

The generated carrier is participants-artifact-access-carrier. Its required relation is write-read-and-custody-relation. The reconstruction must retain authors-version-access-changes-and-disputes-held. Its declared empirical boundary is record-custody-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,

SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-MESSAGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `participants-artifact-access-carrier`.
- **boundary** - which population, place, period and method bound the statement: `record-custody-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `write-read-and-custody-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `authors-version-access-changes-and-disputes-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
participants-artifact-access-carrier__record-custody-bounded__write-read-and-custody-
relation__authors-version-access-changes-and-disputes-held__evidence-and-normative-cl
ass-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-conte
xts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if one party's record becomes uncontested collective memory.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **shared social record**, the consequence is practical and exact: A shared record is a versioned artifact with authorship, access, custody, changes and disputes retained. A scientific claim therefore has to expose the carrier `participants-artifact-access-carrier`, the relation `write-read-and-custody-relation`, the record `authors-version-access-changes-and-disputes-held`, and the boundary `record-custody-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:b662dc9f66a39e517078e882fedbbd96a05b4361aea3be4b7fc90f6d4e70db98`; independent certificate: `sha256:273d068dfdfefb94674679f07f3eeb587aeba13a8fb5095d266dc91dbaede210b`; engine receipt: `sha256:f56e1e73f396479164cdf9c2e4688d84f875d812d4f97c17ce23efbe47ae6ee1`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

11. Common-knowledge boundary

Claim: `SFT-SOCIAL-COMMON-KNOWLEDGE-001`

Family: `observation_action_communication`

Exact statement: Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain.

Forward chain and exact carrier

The generated carrier is `agents-message-receipt-iteration-carrier`. Its required relation is `mutual-observation-depth-relation`. The reconstruction must retain `participants-receipts-depth-and-missing-links-held`. Its declared empirical boundary is `finite-knowledge-depth-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-SHARED-RECORD-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-message-receipt-iteration-carrier.
- **boundary** - which population, place, period and method bound the statement: finite-knowledge-depth-bounded.
- **relation** - which generated interaction or comparison connects the carriers: mutual-observation-depth-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: participants-receipts-depth-and-missing-links-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agents-message-receipt-iteration-carrier__finite-knowledge-depth-bounded__mutual-observation-depth-relation__participants-receipts-depth-and-missing-links-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject completed infinite mutual knowledge or infer it from a public message alone.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **common-knowledge boundary**, the consequence is practical and exact: Common knowledge claims require a declared finite mutual-observation depth and complete receipt chain. A scientific claim therefore has to expose the carrier `agents-message-receipt-iteration-carrier`, the relation `mutual-observation-depth-relation`, the record `participants-receipts-depth-and-missing-links-held`, and the boundary `finite-knowledge-depth-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:71c761cd89d0205dffbc921a7e7697062312d8b84cb6b25b7f0f08a5e0796ef`; independent certificate: `sha256:c4ac04442a2ec7b398bdd2319648c25102ffdb8fce90f7507b6f496d5eb96c35`; engine receipt: `sha256:3b5736ee7fbc85b03e2edad2e4b2a5447fb2baa06d55f9e704e3b01eb099fbd`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

12. Social communication noise

Claim: SFT-SOCIAL-COMMUNICATION-NOISE-001

Family: observation_action_communication

Exact statement: Communication noise retains technical loss, ambiguity, reinterpretation and audience differences.

Forward chain and exact carrier

The generated carrier is message-channel-context-interpretation-carrier. Its required relation is distortion-ambiguity-and-loss-relation. The reconstruction must retain source-versions-audiences-interpretations-and-missingness-held. Its declared empirical boundary is channel-and-audience-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-POPULATION-001, SFT-SOCIAL-COMMON-KNOWLEDGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: message-channel-context-interpretation-carrier.
- **boundary** - which population, place, period and method bound the statement: channel-and-audience-bounded.
- **relation** - which generated interaction or comparison connects the carriers: distortion-ambiguity-and-loss-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: source-versions-audiences-interpretations-and-missingness-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

message-channel-context-interpretation-carrier__channel-and-audience-bounded__distortion-ambiguity-and-loss-relation__source-versions-audiences-interpretations-and-missingness-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if disagreement is automatically deception or if a transformed message is called identical.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 1. Unresolved transports retained: 2. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social communication noise**, the consequence is practical and exact: Communication noise retains technical loss, ambiguity, reinterpretation and audience differences. A scientific claim therefore has to expose the carrier `message-channel-context-interpretation-carrier`, the relation `distortion-ambiguity-and-loss-relation`, the record `source-versions-audiences-interpretations-and-missingness-held`, and the boundary `channel-and-audience-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:e5650bb5680cf1234de2c1018357e47154abe126baf1ad0e5b52e9c2fcf55936; independent certificate: sha256:adfb6dae57e10532692d670594936c5fe7d24ac4154991590103eb8f5784c412; engine receipt: sha256:c609ce1483a76a7a83d8f28c5fd7170640788f6afeff041858b81e20aa23150e. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

13. Coordination

Claim: SFT-SOCIAL-COORDINATION-001

Family: coordination_conflict

Exact statement: Coordination is compatible action around a declared target with dependencies, timing and failures retained.

Forward chain and exact carrier

The generated carrier is agents-actions-joint-target-carrier. Its required relation is timing-and-compatibility-relation. The reconstruction must retain agents-actions-dependencies-time-failures-and-target-held. Its declared empirical boundary is task-and-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-actions-joint-target-carrier.
- **boundary** - which population, place, period and method bound the statement: task-and-context-bounded.
- **relation** - which generated interaction or comparison connects the carriers: timing-and-compatibility-relation.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
agents-actions-dependencies-time-failures-and-target-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
agents-actions-joint-target-carrier__task-and-context-bounded__timing-and-compatibili
ty-relation__agents-actions-dependencies-time-failures-and-target-held__evidence-and-
normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retai
ns-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if simultaneous action proves a shared target or successful coordination proves cooperation.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **coordination**, the consequence is practical and exact: Coordination is compatible action around a declared target with dependencies, timing and failures retained. A scientific claim therefore has to expose the carrier `agents-actions-joint-target-carrier`, the relation `timing-and-compatibility-relation`, the record `agents-actions-dependencies-time-failures-and-target-held`, and the boundary `task-and-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:8ee843badaeca2dffe17716651d75ccdf841d85d07ae08290d49ce1b84645d3c`; independent certificate: `sha256:ce5862b80b25720fd9bb1047054e0cfd2ac7764da3c1167334ca71d426858e4f`; engine receipt: `sha256:332c7b9b943206b9afa4b5b83ce4fc3c08cf81653588e2c217d2966cfc0fcd6d`. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

14. Cooperation

Claim: SFT-SOCIAL-COOPERATION-001

Family: `coordination_conflict`

Exact statement: Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants.

Forward chain and exact carrier

The generated carrier is `agents-contributions-joint-benefit-carrier`. Its required relation is `contribution-benefit-and-reciprocity-relation`. The reconstruction must retain `contributions-costs-benefits-alternatives-time-and-defection-held`. Its declared empirical boundary is `benefit-and-membership-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COORDINATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-contributions-joint-benefit-carrier`.
- **boundary** - which population, place, period and method bound the statement: `benefit-and-membership-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `contribution-benefit-and-reciprocity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `contributions-costs-benefits-alternatives-time-and-defection-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-contributions-joint-benefit-carrier__benefit-and-membership-bounded__contribution-benefit-and-reciprocity-relation__contributions-costs-benefits-alternatives-time-and-defection-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if aggregate gain hides uncompensated loss or coerced contribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **cooperation**, the consequence is practical and exact: Cooperation retains contributions, costs, benefits, alternatives and defection for identified participants. A scientific claim therefore has to expose the carrier agents-contributions-joint-benefit-carrier, the relation contribution-benefit-and-reciprocity-relation, the record contributions-costs-benefits-alternatives-time-and-defection-held, and the boundary benefit-and-membership-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:dcc4601ab91b90960bb694d29ee3f425a79d8f621ec6a36789dc81c4144e624e; independent certificate: sha256:94066ec99ceaa3ad4b9bb8881f2fc59f9008a5bc906070f773c64e196e5bde9d; engine receipt: sha256:5c58d6b54a1282c18b43b1785af6f6f7689a713dee415eb607df14171fc37ea5. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

15. Competition

Claim: SFT-SOCIAL-COMPETITION-001

Family: coordination_conflict

Exact statement: Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained.

Forward chain and exact carrier

The generated carrier is agents-scarce-target-rules-carrier. Its required relation is rival-claim-and-allocation-relation. The reconstruction must retain participants-resource-rules-opportunities-outcomes-and-exclusion-held. Its declared empirical boundary is contest-boundary-declared. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COOPERATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-scarce-target-rules-carrier`.
- **boundary** - which population, place, period and method bound the statement: `contest-boundary-declared`.
- **relation** - which generated interaction or comparison connects the carriers: `rival-claim-and-allocation-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-resource-rules-opportunities-outcomes-and-exclusion-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-scarce-target-rules-carrier__contest-boundary-declared__rival-claim-and-allocation-relation__participants-resource-rules-opportunities-outcomes-and-exclusion-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if unequal access is hidden or all parallel striving is called competition.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **competition**, the consequence is practical and exact: Competition is a rule- and resource-bounded rival claim with opportunities and exclusion retained. A scientific claim therefore has to expose the carrier `agents-scarce-target-rules-carrier`, the relation `rival-claim-and-allocation-relation`, the record `participants-resource-rules-opportunities-outcomes-and-exclusion-held`, and the boundary `contest-boundary-declared` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:85ff9b4dc5a0592bd60ec5457f5d1887210327d677436de07084eb6e53037dfb; independent certificate: sha256:025c6b194e0454aca243c1388f5407a90dadf9183ea76cc2b0f8697af9b3e0de; engine receipt: sha256:6b6d00922b1c86a7c0c38187646d9ec0c62b74075afe7fd0a9b05cc064d6488b. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

16. Conflict

Claim: SFT-SOCIAL-CONFLICT-001

Family: `coordination_conflict`

Exact statement: Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time.

Forward chain and exact carrier

The generated carrier is `agents-incompatible-claims-action-carrier`. Its required relation is `claim-opposition-and-harm-relation`. The reconstruction must retain `claims-capabilities-actions-harms-time-and-alternatives-held`. Its declared empirical boundary is `conflict-definition-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-COMPETITION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-incompatible-claims-action-carrier`.
- **boundary** - which population, place, period and method bound the statement: `conflict-definition-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `claim-opposition-and-harm-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `claims-capabilities-actions-harms-time-and-alternatives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

agents-incompatible-claims-action-carrier__conflict-definition-bounded__claim-opposition-and-harm-relation__claims-capabilities-actions-harms-time-and-alternatives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if disagreement alone becomes violence or one source defines every party's claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **conflict**, the consequence is practical and exact: Conflict retains incompatible claims, capabilities, actions, harms, alternatives and time. A scientific claim therefore has to expose the carrier
agents-incompatible-claims-action-carrier, the relation claim-opposition-and-harm-relation, the record claims-capabilities-actions-harms-time-and-alternatives-held, and the boundary conflict-definition-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:9ebd08ecabf8024f5ef9fdf83370929ddef651fa0d919d8571edd631a1f0f470; independent certificate: sha256:2c810b7e41b0cc35b593d40f4cd19904d1a73bfc411618227df50f53f8ff05c4; engine receipt: sha256:0c2dfd4f8e85ce16ff07f28527f9e08bb0d4677836a294403856f3dc50d04780. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

17. Bargaining

Claim: SFT-SOCIAL-BARGAINING-001

Family: coordination_conflict

Exact statement: Bargaining retains offers, outside options, information, power, time, breakdown and agreement.

Forward chain and exact carrier

The generated carrier is parties-offers-options-agreement-carrier. Its required relation is offer-counteroffer-and-acceptance-relation. The reconstruction must retain options-information-power-time-breakdown-and-outcome-held. Its declared empirical boundary is bargaining-episode-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-CONFLICT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: parties-offers-options-agreement-carrier.
- **boundary** - which population, place, period and method bound the statement: bargaining-episode-bounded.
- **relation** - which generated interaction or comparison connects the carriers: offer-counteroffer-and-acceptance-relation.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
options-information-power-time-breakdown-and-outcome-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
parties-offers-options-agreement-carrier__bargaining-episode-bounded__offer-counterof
fer-and-acceptance-relation__options-information-power-time-breakdown-and-outcome-hel
d__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite
-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if the final agreement proves equal choice or erased alternatives.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does

not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **bargaining**, the consequence is practical and exact: Bargaining retains offers, outside options, information, power, time, breakdown and agreement. A scientific claim therefore has to expose the carrier
parties-offers-options-agreement-carrier, the relation
offer-counteroffer-and-acceptance-relation, the record
options-information-power-time-breakdown-and-outcome-held, and the boundary
bargaining-episode-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:ba594abcde4302b003da60c65d7173467f75f63e1e2b72d60cd8a640758b337d; independent
certificate: sha256:74934c9c5d0c97738e500d38d0ded2577a73903b8b6d62bf9f1ec401ead9266; engine
receipt: sha256:b452e58367879549904f8cc507672a5ef1bbae7e43052b92ec7365f31d52ead9. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

18. Collective action

Claim: SFT-SOCIAL-COLLECTIVE-ACTION-001

Family: coordination_conflict

Exact statement: Collective action retains participation, abstention, contribution, threshold, benefit, failure and context.

Forward chain and exact carrier

The generated carrier is agents-contribution-threshold-outcome-carrier. Its required relation is participation-and-threshold-relation. The reconstruction must retain participants-nonparticipants-costs-benefits-time-and-failures-held. Its declared empirical boundary is action-frame-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COMMUNICATION-NOISE-001, SFT-SOCIAL-BARGAINING-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-contribution-threshold-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `action-frame-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `participation-and-threshold-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `participants-nonparticipants-costs-benefits-time-and-failures-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-contribution-threshold-outcome-carrier__action-frame-bounded__participation-and-threshold-relation__participants-nonparticipants-costs-benefits-time-and-failures-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if outcome visibility erases nonparticipants or failed attempts.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective action**, the consequence is practical and exact: Collective action retains participation, abstention, contribution, threshold, benefit, failure and context. A scientific claim therefore has to expose the carrier
agents-contribution-threshold-outcome-carrier, the relation
participation-and-threshold-relation, the record
participants-nonparticipants-costs-benefits-time-and-failures-held, and the boundary
action-frame-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:31250970e3490d01e0e402efd510a149f19999abe12c57de4b27fa6fecc62508; independent
certificate: sha256:fb3ab7f0710db8a3010a39b62f683894e74c6f3e3f546b8aac8ba588c770e837; engine
receipt: sha256:a7185420f280dd7879b17b1b675bcbd8920bf7d9bcd752b60733ccf468ae2760. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

19. Social norm

Claim: SFT-SOCIAL-NORM-001

Family: `norm_institution_enforcement`

Exact statement: A norm is a context-bound expectation evidenced by behavior and responses to conformity and violation.

Forward chain and exact carrier

The generated carrier is `group-expectation-behaviour-response-carrier`. Its required relation is `expectation-conformity-and-sanction-relation`. The reconstruction must retain
group-context-variation-violations-responses-and-time-held. Its declared empirical boundary is `norm-population-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: group-expectation-behaviour-response-carrier.
- **boundary** - which population, place, period and method bound the statement: norm-population-bounded.
- **relation** - which generated interaction or comparison connects the carriers: expectation-conformity-and-sanction-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: group-context-variation-violations-responses-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
group-expectation-behaviour-response-carrier__norm-population-bounded__expectation-conformity-and-sanction-relation__group-context-variation-violations-responses-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if researcher preference or majority behaviour alone becomes a norm.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd8a85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social norm**, the consequence is practical and exact: A norm is a context-bound expectation evidenced by behaviour and responses to conformity and violation. A scientific claim therefore has to expose the carrier
group-expectation-behaviour-response-carrier, the relation
expectation-conformity-and-sanction-relation, the record
group-context-variation-violations-responses-and-time-held, and the boundary
norm-population-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:cf47c5430671a2ee6d7877ec0a093e0831416bef79d484252710974f8cec2ad8; independent certificate: sha256:c55672eb76bdefc2f65911803490cc866b4b17446ce048dbd955991296ef1b2c; engine receipt: sha256:81ed8141611660faf49244ba772291a6a232970c571f398f99fd19b9bf11bf7b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

20. Social rule

Claim: SFT-SOCIAL-RULE-001

Family: norm_institution_enforcement

Exact statement: A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation.

Forward chain and exact carrier

The generated carrier is issuer-text-scope-subject-carrier. Its required relation is rule-application-relation. The reconstruction must retain authority-text-version-scope-exceptions-and-contestation-held. Its declared empirical boundary is rule-jurisdiction-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-NORM-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: issuer-text-scope-subject-carrier.
- **boundary** - which population, place, period and method bound the statement: rule-jurisdiction-bounded.
- **relation** - which generated interaction or comparison connects the carriers: rule-application-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: authority-text-version-scope-exceptions-and-contestation-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.

- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
issuer-text-scope-subject-carrier__rule-jurisdiction-bounded__rule-application-relati
on__authority-text-version-scope-exceptions-and-contestation-held__evidence-and-norma
tive-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-al
l-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an unwritten inference is presented as a universal rule or version changes disappear.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd8a5e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social rule**, the consequence is practical and exact: A rule retains issuer, text, version, scope, subjects, exceptions, application and contestation. A scientific claim therefore has to expose the carrier issuer-text-scope-subject-carrier, the relation rule-application-relation, the record authority-text-version-scope-exceptions-and-contestation-held, and the boundary rule-jurisdiction-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:1d63d026129a48e15641d759f95b78b90f6fa9cbf102bf7886bb6181f22c45ff; independent certificate: sha256:db327835791bf997a0e987ab45de704aee7fa5d8159d0900863b26fa09f55eaa; engine receipt: sha256:e7ed142368911acc1f16f02f17d142061e03ea2114d5c3afa799e2345403f4b7. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

21. Institution

Claim: SFT-SOCIAL-INSTITUTION-001

Family: norm_institution_enforcement

Exact statement: An institution is a historically repeated governed process, not merely a building, name or consensus.

Forward chain and exact carrier

The generated carrier is roles-rules-practices-resources-carrier. Its required relation is repeated-governed-process-relation. The reconstruction must retain history-membership-rules-resources-compliance-and-change-held. Its declared empirical boundary is institution-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-RULE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `roles-rules-practices-resources-carrier`.
- **boundary** - which population, place, period and method bound the statement: `institution-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `repeated-governed-process-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `history-membership-rules-resources-compliance-and-change-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
roles-rules-practices-resources-carrier__institution-context-bounded__repeated-govern
ed-process-relation__history-membership-rules-resources-compliance-and-change-held__e
vidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-ext
ension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if institutional prestige is used as evidence or if internal conflict is erased.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **institution**, the consequence is practical and exact: An institution is a historically repeated governed process, not merely a building, name or consensus. A scientific claim therefore has to expose the carrier
roles-rules-practices-resources-carrier, the relation repeated-governed-process-relation, the record history-membership-rules-resources-compliance-and-change-held, and the boundary institution-context-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:368c7fa5c901149bd3a9a726af0648c199377412b11bf2b69e1f442255f2bbda; independent certificate: sha256:e81c8c521bf87c0d117c847bf370e1ae7cad09cea9da56a286d1676a1aa505fb; engine receipt: sha256:b9e33b4de5564358ded810f93b3298bea090786567f936270ba09ba3569c9918. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

22. Enforcement

Claim: SFT-SOCIAL-ENFORCEMENT-001

Family: norm_institution_enforcement

Exact statement: Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement.

Forward chain and exact carrier

The generated carrier is rule-monitor-response-subject-carrier. Its required relation is detection-to-response-relation. The reconstruction must retain authority-monitoring-selectivity-process-response-and-appeal-held. Its declared empirical boundary is enforcement-regime-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,

SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-INSTITUTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: rule-monitor-response-subject-carrier.
- **boundary** - which population, place, period and method bound the statement: enforcement-regime-bounded.
- **relation** - which generated interaction or comparison connects the carriers: detection-to-response-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: authority-monitoring-selectivity-process-response-and-appeal-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
rule-monitor-response-subject-carrier__enforcement-regime-bounded__detection-to-response-relation__authority-monitoring-selectivity-process-response-and-appeal-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a written rule proves uniform enforcement or punishment proves legitimacy.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **enforcement**, the consequence is practical and exact: Enforcement retains monitoring, selectivity, procedure, response, appeal and non-enforcement. A scientific claim therefore has to expose the carrier
rule-monitor-response-subject-carrier, the relation detection-to-response-relation, the record
authority-monitoring-selectivity-process-response-and-appeal-held, and the boundary
enforcement-regime-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:e0842d49fdb0a2111b5efa85a79362a99a4bfffecfdec29ea2ac2aceac342dc31; independent
certificate: sha256:f55ac6ca28e96c56b55e84ba317644a10a3f61ac14939cd93a119a14a1cdcd9f; engine
receipt: sha256:b998039d03a7d316d7dbfea6e71a7b83d5a0774f84727d8b47839b3c174a84f2. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

23. Compliance and resistance

Claim: SFT-SOCIAL-COMPLIANCE-001

Family: `norm_institution_enforcement`

Exact statement: Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence.

Forward chain and exact carrier

The generated carrier is `subject-rule-action-reason-carrier`. Its required relation is `rule-to-behaviour-comparison`. The reconstruction must retain `knowledge-capability-motive-action-resistance-and-consequence-held`. Its declared empirical boundary is `compliance-observation-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-COLLECTIVE-ACTION-001`, `SFT-SOCIAL-ENFORCEMENT-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `subject-rule-action-reason-carrier`.
- **boundary** - which population, place, period and method bound the statement: `compliance-observation-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `rule-to-behaviour-comparison`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `knowledge-capability-motive-action-resistance-and-consequence-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
subject-rule-action-reason-carrier__compliance-observation-bounded__rule-to-behaviour-
-comparison__knowledge-capability-motive-action-resistance-and-consequence-held__evid
ence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extens
ion-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report,

model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if observed conformity proves acceptance or nonconformity proves knowledge.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **compliance and resistance**, the consequence is practical and exact: Compliance and resistance remain distinct from ignorance, inability, concealment and coincidence. A scientific claim therefore has to expose the carrier subject-rule-action-reason-carrier, the relation rule-to-behaviour-comparison, the record knowledge-capability-motive-action-resistance-and-consequence-held, and the boundary compliance-observation-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d937c94fc7db7669dc87ad141c82e24bf169bcf5d6e24b02f4d00ee11055503b; independent certificate: sha256:7cad08491ab74f4c15c61894c89a84d98f7cf87a244ba27a461c59cf081156de; engine receipt: sha256:c9e98286efce43ca1d05239813f71d18955c9d9420d1071237c72d6422895580. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

24. Institutional change

Claim: SFT-SOCIAL-INSTITUTIONAL-CHANGE-001

Family: norm_institution_enforcement

Exact statement: Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts.

Forward chain and exact carrier

The generated carrier is institution-prior-event-later-carrier. Its required relation is rule-practice-transition-relation. The reconstruction must retain versions-agents-events-continuities-conflicts-and-time-held. Its declared empirical boundary is historical-transition-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-ACTION-001, SFT-SOCIAL-COMPLIANCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: institution-prior-event-later-carrier.
- **boundary** - which population, place, period and method bound the statement: historical-transition-bounded.
- **relation** - which generated interaction or comparison connects the carriers: rule-practice-transition-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: versions-agents-events-continuities-conflicts-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
institution-prior-event-later-carrier__historical-transition-bounded__rule-practice-t
ransition-relation__versions-agents-events-continuities-conflicts-and-time-held__evid
ence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extens
ion-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a later state is used as its own causal history.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNODC-DATA-001 - transport captured; captured bytes 44411; snapshot
sha256:35af0d6118e639659e259be841e38208a4d0bc13020b971d9ebaa43b41233d57.
- OECD-DATA-EXPLORER-001 - transport captured; captured bytes 173987; snapshot
sha256:cdac198c1032dbb902cd9bfd5bd8659f0a59ab65780aee8c4cd50ad0a24324b5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **institutional change**, the consequence is practical and exact: Institutional change is a source-dated transition retaining continuity, conflict and alternative accounts. A scientific claim therefore has to expose the carrier `institution-prior-event-later-carrier`, the relation `rule-practice-transition-relation`, the record `versions-agents-events-continuities-conflicts-and-time-held`, and the boundary `historical-transition-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:50951d82a43651cc20974ef942b18167fa0f5abf1885ef480a9047c9dad1354f`; independent certificate: `sha256:dc0906bfb8737219ab494d5aa7f0542bff6f847f64b3e08b7d8f882a4d5bcf8a`; engine receipt: `sha256:a3453592fdd413eeda7249449dec8f49cacb169203d83c7888ed55df399c3116`. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

25. Trust relation

Claim: SFT-SOCIAL-TRUST-001

Family: `trust_reputation_knowledge`

Exact statement: Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained.

Forward chain and exact carrier

The generated carrier is `trustor-trustee-vulnerability-expectation-carrier`. Its required relation is `reliance-and-outcome-relation`. The reconstruction must retain `parties-domain-risk-history-action-and-breach-held`. Its declared empirical boundary is `trust-domain-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `trustor-trustee-vulnerability-expectation-carrier`.
- **boundary** - which population, place, period and method bound the statement: `trust-domain-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `reliance-and-outcome-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `parties-domain-risk-history-action-and-breach-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
trustor-trustee-vulnerability-expectation-carrier__trust-domain-bounded__reliance-and-outcome-relation__parties-domain-risk-history-action-and-breach-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if familiarity, approval or one successful outcome proves general trust.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.

- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **trust relation**, the consequence is practical and exact: Trust is domain-specific reliance under vulnerability with history, action, outcome and breach retained. A scientific claim therefore has to expose the carrier trustor-trustee-vulnerability-expectation-carrier, the relation reliance-and-outcome-relation, the record parties-domain-risk-history-action-and-breach-held, and the boundary trust-domain-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:80335b9b7fd6bd22545282ab797c6ff25b598e5539dcfb67ea4b37d8efe3b4c7; independent certificate: sha256:7ba829a0a271c08771d63f39449d2bd15a55eccf6f6789397d3def4af322f0a5; engine receipt: sha256:7d449223848c03df9221d66e6483af9123afd8bf01787c1c24e51500b5d9c9b3. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

26. Reputation

Claim: SFT-SOCIAL-REPUTATION-001

Family: trust_reputation_knowledge

Exact statement: Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible.

Forward chain and exact carrier

The generated carrier is subject-observers-reports-carrier. Its required relation is report-aggregation-and-transmission-relation. The reconstruction must retain sources-audiences-content-time-conflict-and-incentives-held. Its declared empirical boundary is reputation-network-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,

SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-TRUST-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `subject-observers-reports-carrier`.
- **boundary** - which population, place, period and method bound the statement: `reputation-network-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `report-aggregation-and-transmission-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `sources-audiences-content-time-conflict-and-incentives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
subject-observers-reports-carrier__reputation-network-bounded__report-aggregation-and-
-transmission-relation__sources-audiences-content-time-conflict-and-incentives-held__
evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-ex
tension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if popularity or repeated rumor becomes verified character.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot sha256:32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **reputation**, the consequence is practical and exact: Reputation is a network of attributed reports whose sources, incentives, conflicts and audiences remain visible. A scientific claim therefore has to expose the carrier
subject-observers-reports-carrier, the relation
report-aggregation-and-transmission-relation, the record
sources-audiences-content-time-conflict-and-incentives-held, and the boundary
reputation-network-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:1bf73d3007311d972eaf0455d23deb8b1d194f4c82d534794ddb0262445ed717; independent
certificate: sha256:a8f73ab2857155fffb8ba76822bede1a00307ada8fce3235407f057279b4d0df; engine
receipt: sha256:6fac43db587b30f5f66791a478d0c848b583c8dfef18e4ed5b10e16daace7ef9. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

27. Testimony and credibility

Claim: SFT-SOCIAL-TESTIMONY-001

Family: trust_reputation_knowledge

Exact statement: Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution.

Forward chain and exact carrier

The generated carrier is `speaker-claim-audience-evidence-carrier`. Its required relation is `testimony-to-belief-opportunity-relation`. The reconstruction must retain `identity-access-claim-support-conflict-and-reception-held`. Its declared empirical boundary is `testimony-event-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-INSTITUTIONAL-CHANGE-001`, `SFT-SOCIAL-REPUTATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `speaker-claim-audience-evidence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `testimony-event-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `testimony-to-belief-opportunity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `identity-access-claim-support-conflict-and-reception-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`speaker-claim-audience-evidence-carrier__testimony-event-bounded__testimony-to-belief-opportunity-relation__identity-access-claim-support-conflict-and-reception-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if status alone establishes truth or marginal status establishes falsity.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **testimony and credibility**, the consequence is practical and exact: Testimony retains speaker access, claim, support, conflicts, audience and reception without credential substitution. A scientific claim therefore has to expose the carrier `speaker-claim-audience-evidence-carrier`, the relation `testimony-to-belief-opportunity-relation`, the record `identity-access-claim-support-conflict-and-reception-held`, and the boundary `testimony-event-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b6ac1ca14619112a72e8d95677ac9729fd78fa339ca4442ed334ffa727a1b4d1; independent certificate: sha256:22b88834c029aaea55d3e72926f18797abc6543e64db7aefdbc18754c0998942; engine receipt: sha256:4b52d15532ed596150734988a9cfd8ce768f6190152d912d5420a4fb4468c0b5. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

28. Collective knowledge

Claim: SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001

Family: trust_reputation_knowledge

Exact statement: Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained.

Forward chain and exact carrier

The generated carrier is distributed-agents-records-validation-carrier. Its required relation is composition-and-retrieval-relation. The reconstruction must retain contributors-methods-records-disagreements-access-and-time-held. Its declared empirical boundary is knowledge-system-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-TESTIMONY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: distributed-agents-records-validation-carrier.
- **boundary** - which population, place, period and method bound the statement: knowledge-system-bounded.
- **relation** - which generated interaction or comparison connects the carriers: composition-and-retrieval-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: contributors-methods-records-disagreements-access-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
distributed-agents-records-validation-carrier__knowledge-system-bounded__composition-
and-retrieval-relation__contributors-methods-records-disagreements-access-and-time-he
ld__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finit
e-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if consensus vote replaces evidence or inaccessible content is called shared knowledge.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective knowledge**, the consequence is practical and exact: Collective knowledge is distributed, recorded and validated content with contribution, access and disagreement retained. A scientific claim therefore has to expose the carrier distributed-agents-records-validation-carrier, the relation composition-and-retrieval-relation, the record contributors-methods-records-disagreements-access-and-time-held, and the boundary knowledge-system-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d190b023d81a3477be7b1c0f1112755ba3a2f0f9743b9549f8465e2b2d9cb92a; independent certificate: sha256:4a542bc0e9b7626f505f2cb1da3d2550d48034a46bd4499c6e340fdd6848a6a2; engine receipt: sha256:fcefdcf23cfc7590e92f561b84763b94bc6b65e5b19c6efa5a317cd329c2c4af1. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

29. Expertise evidence boundary

Claim: SFT-SOCIAL-EXPERTISE-001

Family: trust_reputation_knowledge

Exact statement: Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token.

Forward chain and exact carrier

The generated carrier is agent-domain-performance-record-carrier. Its required relation is competence-to-task-relation. The reconstruction must retain domain-training-method-performance-errors-and-updates-held. Its declared empirical boundary is expertise-domain-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-COLLECTIVE-KNOWLEDGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agent-domain-performance-record-carrier.
- **boundary** - which population, place, period and method bound the statement: expertise-domain-bounded.
- **relation** - which generated interaction or comparison connects the carriers: competence-to-task-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: domain-training-method-performance-errors-and-updates-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
agent-domain-performance-record-carrier__expertise-domain-bounded__competence-to-task-
-relation__domain-training-method-performance-errors-and-updates-held__evidence-and-n
ormative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retain
s-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject credential, title or confidence as sufficient evidence of a claim.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **expertise evidence boundary**, the consequence is practical and exact: Expertise is domain- and task-bound competence evidenced by methods and performance, not a universal authority token. A scientific claim therefore has to expose the carrier agent-domain-performance-record-carrier, the relation competence-to-task-relation, the record domain-training-method-performance-errors-and-updates-held, and the boundary expertise-domain-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:5426f809ebf39d48907c25cf67c3720b3e9719d4cd9d901486c427d3ee2867f1; independent certificate: sha256:c35c9ba8faeea62b5239f206a61d309b182eb4ef56c97cd994a761a2dd00619a; engine receipt: sha256:bbde304ac7029a5c12d6256e624245f3fbfc8aeb6e6629f72843188ac0541ee0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

30. False and unsupported social information

Claim: SFT-SOCIAL-MISINFORMATION-001

Family: trust_reputation_knowledge

Exact statement: False, unsupported, disputed and corrected claims remain distinct and source-dated.

Forward chain and exact carrier

The generated carrier is claim-source-channel-evidence-carrier. Its required relation is claim-to-support-comparison. The reconstruction must retain versions-sources-support-corrections-reach-and-harm-held. Its declared empirical boundary is claim-and-time-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,

SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-INSTITUTIONAL-CHANGE-001, SFT-SOCIAL-EXPERTISE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `claim-source-channel-evidence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `claim-and-time-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `claim-to-support-comparison`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `versions-sources-support-corrections-reach-and-harm-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
claim-source-channel-evidence-carrier__claim-and-time-bounded__claim-to-support-comparison__versions-sources-support-corrections-reach-and-harm-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if dissent is labelled false by status or if later correction erases prior reach.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot `sha256:32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **false and unsupported social information**, the consequence is practical and exact: False, unsupported, disputed and corrected claims remain distinct and source-dated. A scientific claim therefore has to expose the carrier `claim-source-channel-evidence-carrier`, the relation `claim-to-support-comparison`, the record `versions-sources-support-corrections-reach-and-harm-held`, and the boundary `claim-and-time-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:33987505914509e27797b58608a0edae897eb409b1ba1062a7e1e74df0816580`; independent certificate: `sha256:16004d8efe3b1bdc1b4cdd00ce52f70a10206fc37b64465facdb303488b350e4`; engine receipt: `sha256:8b2d26d118ad33a5180de1fa62f95ba04009100be7cf9f438d2e7f03f3b83f51`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

31. Exchange

Claim: `SFT-SOCIAL-EXCHANGE-001`

Family: `exchange_allocation_distribution`

Exact statement: Exchange retains parties, items, rights, terms, timing, costs and failed transfers.

Forward chain and exact carrier

The generated carrier is `parties-items-rights-transfer-carrier`. Its required relation is `reciprocal-or-directed-transfer-relation`. The reconstruction must retain `parties-items-rights-terms-time-costs-and-failure-held`. Its declared empirical boundary is `exchange-event-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: parties-items-rights-transfer-carrier.
- **boundary** - which population, place, period and method bound the statement: exchange-event-bounded.
- **relation** - which generated interaction or comparison connects the carriers: reciprocal-or-directed-transfer-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: parties-items-rights-terms-time-costs-and-failure-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
parties-items-rights-transfer-carrier__exchange-event-bounded__reciprocal-or-directed-
-transfer-relation__parties-items-rights-terms-time-costs-and-failure-held__evidence-
and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-r
etains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if a price alone describes rights, coercion, quality or distribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- `WORLD-BANK-WDI-001` - transport captured; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- `ILOSTAT-001` - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- `UNSD-DEMOGRAPHIC-001` - transport captured; captured bytes 88501; snapshot `sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **exchange**, the consequence is practical and exact: Exchange retains parties, items, rights, terms, timing, costs and failed transfers. A scientific claim therefore has to expose the carrier `parties-items-rights-transfer-carrier`, the relation `reciprocal-or-directed-transfer-relation`, the record `parties-items-rights-terms-time-costs-and-failure-held`, and the boundary `exchange-event-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:d8e8b8ca8aa35447912697c936ca7dbe3ff3272094698b0138b4d77d48318da7`; independent certificate: `sha256:6033631925cfe63cc88fd9857d98eb4a8225dbaf56d306653a56f6bf42479452`; engine receipt: `sha256:df46ce2287ea5a75408e543fe7a36fd714b95f8fd11c611b97a3d1875c600a58`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

32. Social valuation

Claim: SFT-SOCIAL-VALUE-001

Family: exchange_allocation_distribution

Exact statement: Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration.

Forward chain and exact carrier

The generated carrier is agent-object-purpose-context-carrier. Its required relation is preference-and-tradeoff-relation. The reconstruction must retain agent-purpose-options-information-time-and-conflict-held. Its declared empirical boundary is valuation-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-EXCHANGE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agent-object-purpose-context-carrier.
- **boundary** - which population, place, period and method bound the statement: valuation-context-bounded.
- **relation** - which generated interaction or comparison connects the carriers: preference-and-tradeoff-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: agent-purpose-options-information-time-and-conflict-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agent-object-purpose-context-carrier__valuation-context-bounded__preference-and-trade-off-relation__agent-purpose-options-information-time-and-conflict-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if market price, vote or survey mean is equated with universal human value.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social valuation**, the consequence is practical and exact: Valuation is agent-, purpose-, option- and context-bound and cannot become an intrinsic universal scalar by declaration. A scientific claim therefore has to expose the carrier agent-object-purpose-context-carrier, the relation preference-and-tradeoff-relation, the record agent-purpose-options-information-time-and-conflict-held, and the boundary valuation-context-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and

institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:47babbb5d3db640af04b574c00be5800d752315be65e451c840542df19e2cfe1; independent certificate: sha256:24257056e9390a8c5a061a3c61f9e3650a97b7029afe7f7be7615e7402aff36d; engine receipt: sha256:122e9236d3b52979be9fe329ea942a6fd431eb52b6a405859ce2bd5cf36b5a84. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

33. Resource allocation

Claim: SFT-SOCIAL-ALLOCATION-001

Family: exchange_allocation_distribution

Exact statement: Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses.

Forward chain and exact carrier

The generated carrier is resources-claimants-rule-outcome-carrier. Its required relation is rule-to-distribution-relation. The reconstruction must retain stocks-claims-rule-priority-outcomes-losses-and-time-held. Its declared empirical boundary is allocation-boundary-declared. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-VALUE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: resources-claimants-rule-outcome-carrier.
- **boundary** - which population, place, period and method bound the statement: allocation-boundary-declared.
- **relation** - which generated interaction or comparison connects the carriers: rule-to-distribution-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: stocks-claims-rule-priority-outcomes-losses-and-time-held.

- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
resources-claimants-rule-outcome-carrier__allocation-boundary-declared__rule-to-distribution-relation__stocks-claims-rule-priority-outcomes-losses-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if distributed shares omit the stock, denominator or excluded claimants.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31f3e3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **resource allocation**, the consequence is practical and exact: Allocation retains total resources, claimants, rule, priority, outcome, unmet claims and losses. A scientific claim therefore has to expose the carrier `resources-claimants-rule-outcome-carrier`, the relation `rule-to-distribution-relation`, the record `stocks-claims-rule-priority-outcomes-losses-and-time-held`, and the boundary `allocation-boundary-declared` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

`sha256:d9ddcf56ef58df852c3b6b7836d0974cd0f40e695208ffc4702afd771e7f0519`; independent certificate: `sha256:edd12741785091b149d384f75d5aec2da2a801cff090c8128a248a3670154539`; engine receipt: `sha256:92cc1799ee389a64befd0ba46bc3ff916ead60a0b03f32aecafd9bef4a4c9136`. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

34. Collective production

Claim: SFT-SOCIAL-PRODUCTION-001

Family: `exchange_allocation_distribution`

Exact statement: Production retains labour, material, knowledge, ownership, transformation, output and loss.

Forward chain and exact carrier

The generated carrier is `contributors-input-process-output-carrier`. Its required relation is `input-transformation-output-relation`. The reconstruction must retain `labour-material-knowledge-ownership-losses-and-time-held`. Its declared empirical boundary is `production-process-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-ALLOCATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: contributors-input-process-output-carrier.
- **boundary** - which population, place, period and method bound the statement: production-process-bounded.
- **relation** - which generated interaction or comparison connects the carriers: input-transformation-output-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: labour-material-knowledge-ownership-losses-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
contributors-input-process-output-carrier__production-process-bounded__input-transfor
mation-output-relation__labour-material-knowledge-ownership-losses-and-time-held__evi
dence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-exten
sion-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if output alone erases contributors, unpaid inputs, waste or ownership.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective production**, the consequence is practical and exact: Production retains labour, material, knowledge, ownership, transformation, output and loss. A scientific claim therefore has to expose the carrier contributors-input-process-output-carrier, the relation input-transformation-output-relation, the record labour-material-knowledge-ownership-losses-and-time-held, and the boundary production-process-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:df217c3d6bf052768a084dc7f54301bd2783ad6b9c1c58b7721c1b4e55047102; independent certificate: sha256:fe2865499c449433edea52033b755bfc29b7634789992257111c20d2d6324e59; engine receipt: sha256:ae53c6cd5ece19bc49289d53a331d7283bedc3c36f6b065fe93cdf63aldf41a7. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

35. Distribution and inequality

Claim: SFT-SOCIAL-DISTRIBUTION-001

Family: exchange_allocation_distribution

Exact statement: Distribution retains population, total stock, shares, access, time and missing members.

Forward chain and exact carrier

The generated carrier is population-resource-share-carrier. Its required relation is member-to-share-relation. The reconstruction must retain population-frame-stock-shares-access-time-and-missing-held. Its declared empirical boundary is distribution-population-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,

SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MISINFORMATION-001, SFT-SOCIAL-PRODUCTION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-resource-share-carrier`.
- **boundary** - which population, place, period and method bound the statement: `distribution-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `member-to-share-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `population-frame-stock-shares-access-time-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
population-resource-share-carrier__distribution-population-bounded__member-to-share-r
elation__population-frame-stock-shares-access-time-and-missing-held__evidence-and-nor
mative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-
all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if an average, rank or index replaces the complete share distribution.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **distribution and inequality**, the consequence is practical and exact: Distribution retains population, total stock, shares, access, time and missing members. A scientific claim therefore has to expose the carrier population-resource-share-carrier, the relation member-to-share-relation, the record population-frame-stock-shares-access-time-and-missing-held, and the boundary distribution-population-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:d0ad85eaa66d900468c7e7b34afa33159deb383dcfe5488c216521dbea44ee01; independent
certificate: sha256:6156cda5d927062ab0eb76ec9898486c89c3d6c4fbbd5a8a6f3093f1bebc8caf8c; engine
receipt: sha256:94cd29960f46745e6f2ac0a7dc569aeef6d1918e36572172dfc28005de2bd6a. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

36. Social mobility

Claim: SFT-SOCIAL-MOBILITY-001

Family: exchange_allocation_distribution

Exact statement: Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure.

Forward chain and exact carrier

The generated carrier is `agent-origin-destination-time-carrier`. Its required relation is `state-transition-and-opportunity-relation`. The reconstruction must retain `cohort-frame-origin-destination-path-attrition-and-time-held`. Its declared empirical boundary is `cohort-and-measure-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-MISINFORMATION-001`, `SFT-SOCIAL-DISTRIBUTION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agent-origin-destination-time-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cohort-and-measure-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `state-transition-and-opportunity-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `cohort-frame-origin-destination-path-attrition-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agent-origin-destination-time-carrier__cohort-and-measure-bounded__state-transition-and-opportunity-relation__cohort-frame-origin-destination-path-attrition-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject cross-sectional difference as individual mobility or omit those lost to follow-up.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- ILOSTAT-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social mobility**, the consequence is practical and exact: Mobility is a cohort- and measure-bound transition retaining paths, attrition and opportunity structure. A scientific claim therefore has to expose the carrier
agent-origin-destination-time-carrier, the relation
state-transition-and-opportunity-relation, the record
cohort-frame-origin-destination-path-attrition-and-time-held, and the boundary
cohort-and-measure-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:27674d075ad945e9ed09dcfcfd45533672fcf7379dec377c005f18a96b70d9db; independent certificate: sha256:1089d0b29d4263984680d667e9ab267ad414cd77e8205fed70afe5dd04ee1205; engine receipt: sha256:c5ed76c9b77ad263fb8bd1c66d620aa54b73c373e4fdbb0aa7111d912b219823. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

37. Power relation

Claim: SFT-SOCIAL-POWER-001

Family: power_authority_exclusion

Exact statement: Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance.

Forward chain and exact carrier

The generated carrier is actor-target-capability-dependency-carrier. Its required relation is capacity-to-shape-options-relation. The reconstruction must retain actors-resources-options-dependencies-actions-and-resistance-held. Its declared empirical boundary is power-domain-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: actor-target-capability-dependency-carrier.
- **boundary** - which population, place, period and method bound the statement: power-domain-bounded.
- **relation** - which generated interaction or comparison connects the carriers: capacity-to-shape-options-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: actors-resources-options-dependencies-actions-and-resistance-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
actor-target-capability-dependency-carrier__power-domain-bounded__capacity-to-shape-o
ptions-relation__actors-resources-options-dependencies-actions-and-resistance-held__e
vidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-ext
ension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject status, wealth or outcome alone as complete power evidence.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **power relation**, the consequence is practical and exact: Power is a directional capacity to alter another's available actions, retaining resources, dependency and resistance. A scientific claim therefore has to expose the carrier actor-target-capability-dependency-carrier, the relation capacity-to-shape-options-relation, the record actors-resources-options-dependencies-actions-and-resistance-held, and the boundary power-domain-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:d42e21287c383b6f1aca83483562065d5ba804fb29d1b752f4adab6714c06494; independent certificate: sha256:05a66bce16dd11210d1184b14d825fe09bd3e18a1ff3e7dacc1bfd7da1a7b2be; engine receipt: sha256:4c0a80d595e1a96fcd44f40e8970537a863559f2c0af12e7d1f52726586349f. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

38. Authority

Claim: SFT-SOCIAL-AUTHORITY-001

Family: power_authority_exclusion

Exact statement: Authority is scoped decision capacity with source, procedure, recognition and contestation retained.

Forward chain and exact carrier

The generated carrier is office-subject-rule-recognition-carrier. Its required relation is authorized-decision-relation. The reconstruction must retain source-scope-procedure-recognition-contestation-and-time-held. Its declared empirical boundary is authority-jurisdiction-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-POWER-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: office-subject-rule-recognition-carrier.
- **boundary** - which population, place, period and method bound the statement: authority-jurisdiction-bounded.
- **relation** - which generated interaction or comparison connects the carriers: authorized-decision-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: source-scope-procedure-recognition-contestation-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
office-subject-rule-recognition-carrier__authority-jurisdiction-bounded__authorized-decision-relation__source-scope-procedure-recognition-contestation-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject effective coercion as legitimate authority or formal title as effective control.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.

- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **authority**, the consequence is practical and exact: Authority is scoped decision capacity with source, procedure, recognition and contestation retained. A scientific claim therefore has to expose the carrier office-subject-rule-recognition-carrier, the relation authorized-decision-relation, the record source-scope-procedure-recognition-contestation-and-time-held, and the boundary authority-jurisdiction-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:9f62e64a12d9de1be7460efde638a385261299869188537669c046435db3c6dd; independent certificate: sha256:6d8a0f5be84561352ab211fb21676430c19dc5f33d2d746095e965806432cd74; engine receipt: sha256:0ef12b8ddff302feeee5b78871251d4fc32f15a1a4339cbb8e6276356ffb202c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

39. Social dependency

Claim: SFT-SOCIAL-DEPENDENCY-001

Family: power_authority_exclusion

Exact statement: Dependency retains the needed resource, controller, alternatives, costs and failures.

Forward chain and exact carrier

The generated carrier is agents-needed-resource-alternatives-carrier. Its required relation is access-and-substitution-relation. The reconstruction must retain resource-controller-alternatives-costs-time-and-failures-held. Its declared empirical boundary is dependency-resource-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001,

SFT-SOCIAL-AUTHORITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-needed-resource-alternatives-carrier`.
- **boundary** - which population, place, period and method bound the statement: `dependency-resource-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `access-and-substitution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
`resource-controller-alternatives-costs-time-and-failures-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
`evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
`root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case:
`positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-needed-resource-alternatives-carrier__dependency-resource-bounded__access-and-
substitution-relation__resource-controller-alternatives-costs-time-and-failures-held__
_evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-e
xtension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject repeated interaction or correlation as dependency without the alternative boundary.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31f3ce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social dependency**, the consequence is practical and exact: Dependency retains the needed resource, controller, alternatives, costs and failures. A scientific claim therefore has to expose the carrier agents-needed-resource-alternatives-carrier, the relation access-and-substitution-relation, the record resource-controller-alternatives-costs-time-and-failures-held, and the boundary dependency-resource-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:03ad978cfa831c6fb7b29b76aa1971b2feb47e566d561f15e3559f99976534d9; independent
certificate: sha256:68bf2531feb501fc844780174cab3d40ee9bb234fa4dfa303af3b06a50a69650; engine
receipt: sha256:c1c1151da996efecdc23edb781a9fbaf4077c38bc76b903d61c99218f53f2163. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

40. Exclusion

Claim: SFT-SOCIAL-EXCLUSION-001

Family: power_authority_exclusion

Exact statement: Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases.

Forward chain and exact carrier

The generated carrier is agent-opportunity-gate-decision-carrier. Its required relation is eligibility-access-and-denial-relation. The reconstruction must retain criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held. Its declared empirical boundary is access-system-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001,
SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
 SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
 SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
 SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,
 SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001,
 SFT-SOCIAL-DEPENDENCY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agent-opportunity-gate-decision-carrier.
- **boundary** - which population, place, period and method bound the statement: access-system-bounded.
- **relation** - which generated interaction or comparison connects the carriers: eligibility-access-and-denial-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agent-opportunity-gate-decision-carrier__access-system-bounded__eligibility-access-and-denial-relation__criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject if only successful applicants define access or if absence from a system proves free choice.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31f3ce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **exclusion**, the consequence is practical and exact: Exclusion retains opportunity, eligibility, gate, decision, appeal and unobserved discouraged cases. A scientific claim therefore has to expose the carrier agent-opportunity-gate-decision-carrier, the relation eligibility-access-and-denial-relation, the record criteria-gatekeeper-process-outcome-appeal-and-hidden-cases-held, and the boundary access-system-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:02341d7fa6241cee664c3be50f2ed4dbad68b9fcbbeef88a39704a55442e22bc3; independent certificate: sha256:969ca761928a47b3d651667c6330dad3ddfd8c8dc0bac1cf09fd906e2e12acd5; engine receipt: sha256:a706086b718bc0c37342a9586801a6b6e53528d197bbd7e592a47249c98d23b4. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

41. Stratification

Claim: SFT-SOCIAL-STRATIFICATION-001

Family: power_authority_exclusion

Exact statement: Stratification is persistent ordered access across a declared population and resource set with mobility retained.

Forward chain and exact carrier

The generated carrier is population-positions-resources-relations-carrier. Its required relation is ordered-access-and-reproduction-relation. The reconstruction must retain frame-groups-resources-mobility-institutions-and-time-held. Its declared empirical boundary is stratification-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-EXCLUSION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: population-positions-resources-relations-carrier.
- **boundary** - which population, place, period and method bound the statement: stratification-context-bounded.
- **relation** - which generated interaction or comparison connects the carriers: ordered-access-and-reproduction-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: frame-groups-resources-mobility-institutions-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

population-positions-resources-relations-carrier__stratification-context-bounded__ordered-access-and-reproduction-relation__frame-groups-resources-mobility-institutions-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject universal hierarchy from one measure, institution or historical context.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd8a5e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **stratification**, the consequence is practical and exact: Stratification is persistent ordered access across a declared population and resource set with mobility retained. A scientific claim therefore has to expose the carrier

population-positions-resources-relations-carrier, the relation
ordered-access-and-reproduction-relation, the record
frame-groups-resources-mobility-institutions-and-time-held, and the boundary
stratification-context-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:0bc297cd2916ec76cd06445d82a2034411a22a4c102f72082a2cbb15f14aa16; independent certificate: sha256:1d3743391415860558b3ba7e53ce11d62175096f48a1f95ed3ac42be127b07c7; engine receipt: sha256:bdace614acbf29d3c8007db7999b6fdfe0cf4079dd7c93d4785cc2aa5127a5da. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

42. Scientific and institutional gatekeeping

Claim: SFT-SOCIAL-GATEKEEPING-001

Family: power_authority_exclusion

Exact statement: Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion.

Forward chain and exact carrier

The generated carrier is contributor-gate-resource-decision-carrier. Its required relation is access-selection-and-recognition-relation. The reconstruction must retain criteria-costs-credentials-decisions-appeals-and-lost-contributions-held. Its declared empirical boundary is institution-and-period-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-MOBILITY-001, SFT-SOCIAL-STRATIFICATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: contributor-gate-resource-decision-carrier.
- **boundary** - which population, place, period and method bound the statement: institution-and-period-bounded.
- **relation** - which generated interaction or comparison connects the carriers: access-selection-and-recognition-relation.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
criteria-costs-credentials-decisions-appeals-and-lost-contributions-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
contributor-gate-resource-decision-carrier__institution-and-period-bounded__access-se
lection-and-recognition-relation__criteria-costs-credentials-decisions-appeals-and-lo
st-contributions-held__evidence-and-normative-class-explicit__root-bound-forward-forc
ing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject prestige as scientific evidence or criticism of prestige as automatic admission of a claim.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd8a5e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- UNHCR-REFUGEE-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UNSD-DEMOGRAPHIC-001 - transport captured; captured bytes 88501; snapshot
sha256:cc76514a31fce3c8bbc7a37068be70d586fc848cc37051539f736c9fdc0d9ddb.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **scientific and institutional gatekeeping**, the consequence is practical and exact: Gatekeeping evidence retains criteria, costs, credentials, decisions, appeals and contributions made invisible by exclusion. A scientific claim therefore has to expose the carrier contributor-gate-resource-decision-carrier, the relation access-selection-and-recognition-relation, the record criteria-costs-credentials-decisions-appeals-and-lost-contributions-held, and the boundary institution-and-period-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 7d165494602d39b9e70ad90e6115c4ce8f177a3abab13ed4f1af1eb1128266ac; independent certificate: sha256: 32e7f28723577ddba3fb1c574fd2b654e273181f9c6e73ad16b2d0dbbc00e362; engine receipt: sha256: 9d1f3fb1cc3eb7127d43c77fd7f279c9f22c1c8f02e50075e4aae47f0589b4c2. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

43. Social network

Claim: SFT-SOCIAL-SOCIAL-NETWORK-001

Family: network_diffusion_transition

Exact statement: A social network retains node identity, link meaning, direction, time, sampling and missing links.

Forward chain and exact carrier

The generated carrier is agents-typed-links-time-carrier. Its required relation is graph-and-interaction-relation. The reconstruction must retain nodes-link-types-direction-weights-time-and-missing-held. Its declared empirical boundary is network-construction-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-typed-links-time-carrier`.
- **boundary** - which population, place, period and method bound the statement: `network-construction-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `graph-and-interaction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `nodes-link-types-direction-weights-time-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
agents-typed-links-time-carrier__network-construction-bounded__graph-and-interaction-
relation__nodes-link-types-direction-weights-time-and-missing-held__evidence-and-norm
ative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-a
ll-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject an abstract edge as a social tie without measurement or infer absent links outside the sample.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social network**, the consequence is practical and exact: A social network retains node identity, link meaning, direction, time, sampling and missing links. A scientific claim therefore has to expose the carrier `agents-typed-links-time-carrier`, the relation `graph-and-interaction-relation`, the record `nodes-link-types-direction-weights-time-and-missing-held`, and the boundary `network-construction-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:
sha256:31009db1a0f6ef2e5446e5a37ce6c1f8fc68e72f194e00817b46d23364e4c963; independent
certificate: sha256:4a76d92075babced03efcce72fedfe449ca485be46ce90f02ff6a29a3e9e7d77; engine
receipt: sha256:1d731ec9187e36749fd901e73d419bf40d6f0c5757d5b5db3d97b44eab87c5bc. The
independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

44. Social diffusion

Claim: SFT-SOCIAL-DIFFUSION-001

Family: `network_diffusion_transition`

Exact statement: Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs.

Forward chain and exact carrier

The generated carrier is `content-agents-links-adoptions-carrier`. Its required relation is `exposure-transmission-adoption-relation`. The reconstruction must retain `source-path-opportunity-time-nonadoption-and-external-input-held`. Its declared empirical boundary is `diffusion-process-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,

SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-SOCIAL-NETWORK-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `content-agents-links-adoptions-carrier`.
- **boundary** - which population, place, period and method bound the statement: `diffusion-process-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `exposure-transmission-adoption-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `source-path-opportunity-time-nonadoption-and-external-input-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
content-agents-links-adoptions-carrier__diffusion-process-bounded__exposure-transmission-adoption-relation__source-path-opportunity-time-nonadoption-and-external-input-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject temporal clustering as peer transmission or exposure as adoption.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social diffusion**, the consequence is practical and exact: Diffusion retains source, path, exposure opportunity, adoption, nonadoption and external inputs. A scientific claim therefore has to expose the carrier
content-agents-links-adoptions-carrier, the relation
exposure-transmission-adoption-relation, the record
source-path-opportunity-time-nonadoption-and-external-input-held, and the boundary
diffusion-process-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:c1c60d5b76ac2b904a34d662bc966cd5f3d0840c70b5793b33b073e036677666; independent
certificate: sha256:915a342c552e10508f6482be5901b42ec4249ea04eed2bf04bf476164928f666; engine
receipt: sha256:67f740214a3cb3f17d5980370451e1a0229775652bfb7552b53a50c04fd9708b. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

45. Collective synchronization

Claim: SFT-SOCIAL-SYNCHRONIZATION-001

Family: `network_diffusion_transition`

Exact statement: Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained.

Forward chain and exact carrier

The generated carrier is `agents-rhythms-coupling-carrier`. Its required relation is `phase-recurrence-and-coupling-relation`. The reconstruction must retain `states-times-links-lags-breaks-and-controls-held`. Its declared empirical boundary is `synchrony-window-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-GATEKEEPING-001`, `SFT-SOCIAL-DIFFUSION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `agents-rhythms-coupling-carrier`.
- **boundary** - which population, place, period and method bound the statement: `synchrony-window-bounded`.
- **relation** - which generated interaction or comparison connects the carriers:
`phase-recurrence-and-coupling-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
`states-times-links-lags-breaks-and-controls-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
`evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
`root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case:
`positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`agents-rhythms-coupling-carrier__synchrony-window-bounded__phase-recurrence-and-coupling-relation__states-times-links-lags-breaks-and-controls-held__evidence-and-normative`

```
e-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-c
ontexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject coincident peaks or shared schedules as endogenous synchronization.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective synchronization**, the consequence is practical and exact. Synchronization is recurrent phase relation with coupling, lag, breaks and common-cause controls retained. A scientific claim therefore has to expose the carrier `agents-rhythms-coupling-carrier`, the relation `phase-recurrence-and-coupling-relation`, the record `states-times-links-lags-breaks-and-controls-held`, and the boundary `synchrony-window-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a

replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:cb4ac5c069322e98e98291bb2fee7ba9f9695b6fd4aa44ca872fa1c56643b3ad; independent certificate: sha256:f93206d0491dbd6dbc5bec41de066242248008603d5a3ff5e408590a68a8ad86; engine receipt: sha256:80e44dc1ceefce9f9a613530d8d23f3772b0bb3748587e4035c9302b56b8d294. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

46. Collective cascade

Claim: SFT-SOCIAL-CASCADE-001

Family: network_diffusion_transition

Exact statement: A cascade retains ordered local transitions, paths, nontransitions and external causes.

Forward chain and exact carrier

The generated carrier is agents-state-threshold-path-carrier. Its required relation is local-transition-propagation-relation. The reconstruction must retain order-links-opportunities-nontransitions-and-boundaries-held. Its declared empirical boundary is cascade-event-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-SYNCHRONIZATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-state-threshold-path-carrier.
- **boundary** - which population, place, period and method bound the statement: cascade-event-bounded.
- **relation** - which generated interaction or comparison connects the carriers: local-transition-propagation-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: order-links-opportunities-nontransitions-and-boundaries-held.

- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
agents-state-threshold-path-carrier__cascade-event-bounded__local-transition-propagation-relation__order-links-opportunities-nontransitions-and-boundaries-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject aggregate change without an observed transmission chain as a cascade.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective cascade**, the consequence is practical and exact: A cascade retains ordered local transitions, paths, nontransitions and external causes. A scientific claim therefore has to expose the carrier agents-state-threshold-path-carrier, the relation local-transition-propagation-relation, the record order-links-opportunities-nontransitions-and-boundaries-held, and the boundary cascade-event-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 4f03311bc088ecf179f3cdd1999442e4fd5643fce0c3789a78213a62f9b00ed1; independent certificate: sha256: 703a8bec113535408ad2ab82d4a78bc2b5da0ea97fd4c1869fe5e8d5ab2c7696; engine receipt: sha256: 217adf4d391053fa69d18f11faa90e665f18c827863e7eb1bb41ad447b770c03. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

47. Collective transition

Claim: SFT-SOCIAL-COLLECTIVE-TRANSITION-001

Family: network_diffusion_transition

Exact statement: Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained.

Forward chain and exact carrier

The generated carrier is population-prior-event-later-carrier. Its required relation is distribution-change-and-recurrence-relation. The reconstruction must retain frame-states-time-drivers-controls-and-reversals-held. Its declared empirical boundary is transition-population-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-CASCADE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-prior-event-later-carrier`.
- **boundary** - which population, place, period and method bound the statement: `transition-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `distribution-change-and-recurrence-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `frame-states-time-drivers-controls-and-reversals-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
population-prior-event-later-carrier__transition-population-bounded__distribution-change-and-recurrence-relation__frame-states-time-drivers-controls-and-reversals-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject one threshold or Fold coordinate as a universal physical social tipping value.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective transition**, the consequence is practical and exact: Collective transition is a population-distribution change with drivers, controls, reversals and historical boundary retained. A scientific claim therefore has to expose the carrier population-prior-event-later-carrier, the relation distribution-change-and-recurrence-relation, the record frame-states-time-drivers-controls-and-reversals-held, and the boundary transition-population-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:5344fb8c677fa6e2ec536a88efb972600c2c862bd1e971d88a4b7cddd46e1a58; independent certificate: sha256:486fb98304fe2b574df95dcebbab89d10c615c01cf6ef007758633897b2dee2e; engine receipt: sha256:2d9c7a6452a51c3691d4ff3f6db6294a483b0723fd357c8e21430ae1a474f510. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

48. Socio-economic cycle

Claim: SFT-SOCIAL-SOCIAL-CYCLE-001

Family: network_diffusion_transition

Exact statement: A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change.

Forward chain and exact carrier

The generated carrier is time-indexed-collective-state-carrier. Its required relation is recurrence-and-feedback-relation. The reconstruction must retain variables-frame-phases-shocks-breaks-and-time-held. Its declared empirical boundary is cycle-series-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,

SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-GATEKEEPING-001, SFT-SOCIAL-COLLECTIVE-TRANSITION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `time-indexed-collective-state-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cycle-series-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `recurrence-and-feedback-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `variables-frame-phases-shocks-breaks-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
time-indexed-collective-state-carrier__cycle-series-bounded__recurrence-and-feedback-
relation__variables-frame-phases-shocks-breaks-and-time-held__evidence-and-normative-
class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-con
texts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject two peaks or a narrative phase label as a universal periodic law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor

- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1.
The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or
prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation:
false.

- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **socio-economic cycle**, the consequence is practical and exact: A social cycle requires recurrence under a declared state definition and retains shocks, broken cycles and measurement change. A scientific claim therefore has to expose the carrier time-indexed-collective-state-carrier, the relation recurrence-and-feedback-relation, the record variables-frame-phases-shocks-breaks-and-time-held, and the boundary cycle-series-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:da80c3e0458dd776425d3b4740e8bcde870bf4e71917038140cf38dfeeea9d24; independent
certificate: sha256:e7982b5485819f0505afe625df8df6a7557cb08a1aba4940183fcel3aadaa815; engine
receipt: sha256:9959bc1bdebd09230f2219c1b2122637585365e4eca719e1e11ff44ecfbba248. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

49. Culture

Claim: SFT-SOCIAL-CULTURE-001

Family: culture_language_transmission

Exact statement: Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group.

Forward chain and exact carrier

The generated carrier is `practices-symbols-groups-history-carrier`. Its required relation is `transmission-variation-and-reproduction-relation`. The reconstruction must retain `members-practices-meanings-variation-sources-and-time-held`. Its declared empirical boundary is `cultural-context-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-SOCIAL-CYCLE-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `practices-symbols-groups-history-carrier`.
- **boundary** - which population, place, period and method bound the statement: `cultural-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `transmission-variation-and-reproduction-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `members-practices-meanings-variation-sources-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`practices-symbols-groups-history-carrier__cultural-context-bounded__transmission-variation-and-reproduction-relation__members-practices-meanings-variation-sources-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject stereotype, majority report or one observer's taxonomy as complete culture.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot
sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **culture**, the consequence is practical and exact: Culture is historically transmitted and varied practice and meaning, not an immutable property of a labelled group. A scientific claim therefore has to expose the carrier
practices-symbols-groups-history-carrier, the relation
transmission-variation-and-reproduction-relation, the record
members-practices-meanings-variation-sources-and-time-held, and the boundary
cultural-context-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:00eec0ddc4edfac7b41c078511ada7177ca3504c557ed711c0f0fbd83c271dae; independent certificate: sha256:7331bb3d2844eb8686d1aacad8647bc4baf9c16b6081f273f5ef8ffffee1cc7e; engine receipt: sha256:e80d173192171fcbce88fd26c3c57e7a3191d5ed3d4ca98db62ff54531705b593. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

50. Social symbol and meaning

Claim: SFT-SOCIAL-SYMBOL-001

Family: culture_language_transmission

Exact statement: A symbol's social meaning is user-, context- and history-bound with alternatives retained.

Forward chain and exact carrier

The generated carrier is form-users-context-interpretation-carrier. Its required relation is form-to-meaning-relation. The reconstruction must retain form-users-context-alternatives-history-and-change-held. Its declared empirical boundary is interpretive-community-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-CULTURE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: form-users-context-interpretation-carrier.
- **boundary** - which population, place, period and method bound the statement: interpretive-community-bounded.
- **relation** - which generated interaction or comparison connects the carriers: form-to-meaning-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: form-users-context-alternatives-history-and-change-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.

- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
form-users-context-interpretation-carrier__interpretive-community-bounded__form-to-meaning-relation__form-users-context-alternatives-history-and-change-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject a form as carrying one universal social meaning without a community bridge.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot
sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social symbol and meaning**, the consequence is practical and exact: A symbol's social meaning is user-, context- and history-bound with alternatives retained. A scientific claim therefore has to expose the carrier form-users-context-interpretation-carrier, the relation form-to-meaning-relation, the record form-users-context-alternatives-history-and-change-held, and the boundary interpretive-community-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 6ea677625fe5f17f1e4ba936ef4dc0fb359ba42b3a669b60c689db807d349f5e; independent certificate: sha256: 878fe2b2d32a0fa191bf64ef45c8a4dd2156622fa35c0daea2b31cec00717a53; engine receipt: sha256: 2954e69231b375ec0440a69da946ab51192107b05360efbbafe3eba1debc9f84. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

51. Language community

Claim: SFT-SOCIAL-LANGUAGE-COMMUNITY-001

Family: culture_language_transmission

Exact statement: A language community retains speakers, varieties, contexts, power, change and corpus limits.

Forward chain and exact carrier

The generated carrier is speakers-practices-varieties-interactions-carrier. Its required relation is linguistic-transmission-and-coordination-relation. The reconstruction must retain speakers-varieties-context-power-change-and-access-held. Its declared empirical boundary is community-and-corpus-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-SYMBOL-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `speakers-practices-varieties-interactions-carrier`.
- **boundary** - which population, place, period and method bound the statement: `community-and-corpus-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `linguistic-transmission-and-coordination-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `speakers-varieties-context-power-change-and-access-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
speakers-practices-varieties-interactions-carrier__community-and-corpus-bounded__linguistic-transmission-and-coordination-relation__speakers-varieties-context-power-change-and-access-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject standard variety, dictionary or state boundary as the whole language community.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **language community**, the consequence is practical and exact: A language community retains speakers, varieties, contexts, power, change and corpus limits. A scientific claim therefore has to expose the carrier speakers-practices-varieties-interactions-carrier, the relation linguistic-transmission-and-coordination-relation, the record speakers-varieties-context-power-change-and-access-held, and the boundary community-and-corpus-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:a2449c1a58553ce22bfe461991debe7d50416a2c73b03c1aa2dec6a39eed66d4; independent certificate: sha256:f6ec7cc45e1d644c0eb2c8badce0e6a5ec28a36dd480c65beca1547f6ea41cfe; engine receipt: sha256:27f3aec512e0ec3ae29a7826f51de3f4676d678d60865565dcf1e3ea595a536c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

52. Education and knowledge transmission

Claim: SFT-SOCIAL-EDUCATION-001

Family: culture_language_transmission

Exact statement: Education evidence retains access, instruction, participation, assessment, attrition and context.

Forward chain and exact carrier

The generated carrier is learners-teachers-content-institution-carrier. Its required relation is instruction-participation-learning-relation. The reconstruction must retain access-method-content-assessment-attrition-context-and-time-held. Its declared empirical boundary is education-setting-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,

SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-LANGUAGE-COMMUNITY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: learners-teachers-content-institution-carrier.
- **boundary** - which population, place, period and method bound the statement: education-setting-bounded.
- **relation** - which generated interaction or comparison connects the carriers: instruction-participation-learning-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: access-method-content-assessment-attribution-context-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
learners-teachers-content-institution-carrier__education-setting-bounded__instruction-participation-learning-relation__access-method-content-assessment-attribution-context-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject credential attainment as complete learning or nonattendance as inability.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor

- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1.
The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or
prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation:
false.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot
sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot
sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and
preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **education and knowledge transmission**, the consequence is practical and exact: Education evidence retains access, instruction, participation, assessment, attrition and context. A scientific claim therefore has to expose the carrier
learners-teachers-content-institution-carrier, the relation
instruction-participation-learning-relation, the record
access-method-content-assessment-attrition-context-and-time-held, and the boundary
education-setting-bounded together. If any one is hidden, the result may remain a description, interpretation, model,
political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:fcdbc4ac46e81898a804d663fb56755e5381c3562bc06677cbda0f7d0ee89d25; independent
certificate: sha256:10184ba80f4dbbd8bdb5ad7134820662ef01ff47543cad6dd6cfbe86f6e3004; engine
receipt: sha256:c3e3422783196463271d69df0f596ae0d30a3b0485b0ba8f0723b8787e235e0d. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

53. Media system

Claim: SFT-SOCIAL-MEDIA-001

Family: culture_language_transmission

Exact statement: A media system retains ownership, production, selection, distribution, reach, response and missing exposure.

Forward chain and exact carrier

The generated carrier is `producers-content-platform-audience-carrier`. Its required relation is `production-selection-distribution-relation`. The reconstruction must retain `ownership-algorithms-content-reach-responses-and-missing-held`. Its declared empirical boundary is `media-system-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-SOCIAL-CYCLE-001`, `SFT-SOCIAL-EDUCATION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `producers-content-platform-audience-carrier`.
- **boundary** - which population, place, period and method bound the statement: `media-system-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `production-selection-distribution-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `ownership-algorithms-content-reach-responses-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`producers-content-platform-audience-carrier__media-system-bounded__production-selection-distribution-relation__ownership-algorithms-content-reach-responses-and-missing-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject published content as received information or platform metrics as all social response.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot `sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d`.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot `sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1`.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **media system**, the consequence is practical and exact: A media system retains ownership, production, selection, distribution, reach, response and missing exposure. A scientific claim therefore has to expose the carrier `producers-content-platform-audience-carrier`, the relation `production-selection-distribution-relation`, the record `ownership-algorithms-content-reach-responses-and-missing-held`, and the boundary `media-system-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 705a33e6d16c00e608670dacf06b9ec4bb644df9541bfe64d06b044fb60efb58; independent certificate: sha256: 30437a814bb684453f210b41a9118d0a779e9dbfbae909f1206e41834595c563; engine receipt: sha256: d746622ff5a470f4d18e70033d6bbf1a9ca1377736df7efca8193c286527fbe1. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

54. Collective memory

Claim: SFT-SOCIAL-COLLECTIVE-MEMORY-001

Family: culture_language_transmission

Exact statement: Collective memory is a contested transmission of records, narratives and practices with omissions retained.

Forward chain and exact carrier

The generated carrier is groups-records-narratives-practices-carrier. Its required relation is selection-transmission-and-contestation-relation. The reconstruction must retain sources-custody-omissions-versions-contestation-and-time-held. Its declared empirical boundary is memory-community-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-SOCIAL-CYCLE-001, SFT-SOCIAL-MEDIA-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: groups-records-narratives-practices-carrier.
- **boundary** - which population, place, period and method bound the statement: memory-community-bounded.
- **relation** - which generated interaction or comparison connects the carriers: selection-transmission-and-contestation-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: sources-custody-omissions-versions-contestation-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
groups-records-narratives-practices-carrier__memory-community-bounded__selection-transmission-and-contestation-relation__sources-custody-omissions-versions-contestation-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject dominant narrative as complete history or a destroyed record as evidence of no event.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UNESCO-UIS-001 - transport captured; captured bytes 112727; snapshot sha256:030cd7efb36181c78ef896187dac1fff5d72104475cd2e2376dd6fc65d82335d.
- GSS-NORC-001 - transport captured; captured bytes 67309; snapshot sha256:f6930ad92cbddc1db336550b601218344c2f948a44996a3f1176fc287cbfe7e1.
- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective memory**, the consequence is practical and exact: Collective memory is a contested transmission of records, narratives and practices with omissions retained. A scientific claim therefore has to expose the carrier groups-records-narratives-practices-carrier, the relation selection-transmission-and-contestation-relation, the record sources-custody-omissions-versions-contestation-and-time-held, and the boundary memory-community-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:81f760e248518bd7808ff4361d4be0a4a895477a1b6d168860b8c683c62b0142; independent certificate: sha256:e80d827e7845e127ba3bccd64d8ba48c5ce0b3c4e3ce173f5ffe6213a410c36a; engine receipt: sha256:787f1074b105c114d5e3506c01c9a626a39f67ac8a73f37ee6f75893632723e2. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

55. Individual-to-collective composition

Claim: SFT-SOCIAL-MICRO-MACRO-001

Family: multi_level_causation

Exact statement: Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions.

Forward chain and exact carrier

The generated carrier is agents-relations-aggregate-carrier. Its required relation is composition-and-distribution-relation. The reconstruction must retain agent-states-links-weights-aggregation-loss-and-time-held. Its declared empirical boundary is composition-rule-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-relations-aggregate-carrier.
- **boundary** - which population, place, period and method bound the statement: composition-rule-bounded.
- **relation** - which generated interaction or comparison connects the carriers: composition-and-distribution-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: agent-states-links-weights-aggregation-loss-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
agents-relations-aggregate-carrier__composition-rule-bounded__composition-and-distribution-relation__agent-states-links-weights-aggregation-loss-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject an average as a causal collective agent or infer each individual from the aggregate.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **individual-to-collective composition**, the consequence is practical and exact: Individual states compose a collective only through an explicit relation and aggregation rule retaining lost distinctions. A scientific claim therefore has to expose the carrier agents-relations-aggregate-carrier, the relation composition-and-distribution-relation, the record agent-states-links-weights-aggregation-loss-and-time-held, and the boundary composition-rule-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:2994882565a7ebc4b98a68483553bf9bb856ba11d7edfb7f458a4b15cc64dabb; independent certificate: sha256:0c68bd341fb81524da139fcf7628e5ccd170394da760e639cf88843ceda5d24a; engine receipt: sha256:cff9adcd8d1aa01d9fe52c2fe7b60bd3275ac72a813de37b7ffcc9cf2ae99eb0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

56. Collective-to-individual constraint

Claim: SFT-SOCIAL-MACRO-MICRO-001

Family: multi_level_causation

Exact statement: Collective structures affect individuals through observable rules, resources, exposure and expectations.

Forward chain and exact carrier

The generated carrier is institution-agent-options-action-carrier. Its required relation is rule-resource-and-expectation-constraint-relation. The reconstruction must retain institution-options-exposure-capability-response-and-time-held. Its declared empirical boundary is constraint-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001,

SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001,
 SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001,
 SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001,
 SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001,
 SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001,
 SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001,
 SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,
 SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,
 SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001,
 SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001,
 SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MICRO-MACRO-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: institution-agent-options-action-carrier.
- **boundary** - which population, place, period and method bound the statement: constraint-context-bounded.
- **relation** - which generated interaction or comparison connects the carriers:
rule-resource-and-expectation-constraint-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
institution-options-exposure-capability-response-and-time-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
institution-agent-options-action-carrier__constraint-context-bounded__rule-resource-a
nd-expectation-constraint-relation__institution-options-exposure-capability-response-
and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__pos
itive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject a macro label as direct cause without a transmission path.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective-to-individual constraint**, the consequence is practical and exact: Collective structures affect individuals through observable rules, resources, exposure and expectations. A scientific claim therefore has to expose the carrier `institution-agent-options-action-carrier`, the relation `rule-resource-and-expectation-constraint-relation`, the record `institution-options-exposure-capability-response-and-time-held`, and the boundary `constraint-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b3950fa9b3e081ec0c83406b18c7b85c3eead79f0b85cabfc53ee0820d0a576b; independent certificate: sha256:d5c2e489201f063085428ae65370cf714079355f0bd5d5bad2929120d359609b; engine receipt: sha256:f67cabe097f5789ea4637b873b80b854246c72619c73ef360a3cf62538e7f06b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

57. Collective emergence

Claim: SFT-SOCIAL-EMERGENCE-001

Family: multi_level_causation

Exact statement: An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained.

Forward chain and exact carrier

The generated carrier is agents-relations-new-collective-property-carrier. Its required relation is composition-and-nonidentity-relation. The reconstruction must retain components-links-property-tests-alternatives-and-scale-held. Its declared empirical boundary is emergence-definition-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MACRO-MICRO-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: agents-relations-new-collective-property-carrier.
- **boundary** - which population, place, period and method bound the statement: emergence-definition-bounded.
- **relation** - which generated interaction or comparison connects the carriers: composition-and-nonidentity-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: components-links-property-tests-alternatives-and-scale-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

agents-relations-new-collective-property-carrier__emergence-definition-bounded__composition-and-nonidentity-relation__components-links-property-tests-alternatives-and-scale-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject complexity language as proof or erase component contributions.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot `sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927`.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot `sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f`.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective emergence**, the consequence is practical and exact: An emergent collective property is composition-dependent and not identical to any one component, with tests and scale retained. A scientific claim therefore has to expose the carrier `agents-relations-new-collective-property-carrier`, the relation `composition-and-nonidentity-relation`, the record `components-links-property-tests-alternatives-and-scale-held`, and the boundary `emergence-definition-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:3f76a9f8ee810f734a9f1478a5c24b937eb59fd319e28dd60d32679fd229f995; independent certificate: sha256:2efacbf799260066515864ba843bfe383153f5b52affc4b72d9f603a5fe90aae; engine receipt: sha256:539df59af1716049ceaaec48ce23f8df321d8041e8ce99f262e5517c81ea252c. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

58. Human and artificial multi-agent collective

Claim: SFT-SOCIAL-MULTI-AGENT-001

Family: multi_level_causation

Exact statement: A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight.

Forward chain and exact carrier

The generated carrier is heterogeneous-agents-protocol-environment-carrier. Its required relation is interaction-adaptation-and-outcome-relation. The reconstruction must retain agent-types-rules-resources-actions-failures-and-oversight-held. Its declared empirical boundary is system-and-deployment-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-EMERGENCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: heterogeneous-agents-protocol-environment-carrier.
- **boundary** - which population, place, period and method bound the statement: system-and-deployment-bounded.
- **relation** - which generated interaction or comparison connects the carriers: interaction-adaptation-and-outcome-relation.

- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
agent-types-rules-resources-actions-failures-and-oversight-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
heterogeneous-agents-protocol-environment-carrier__system-and-deployment-bounded__interaction-adaptation-and-outcome-relation__agent-types-rules-resources-actions-failures-and-oversight-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject simulated coordination as real-world governance evidence or opaque agents as lawful social causes.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **human and artificial multi-agent collective**, the consequence is practical and exact: A mixed collective retains human and artificial agent types, protocols, resources, failures and oversight. A scientific claim therefore has to expose the carrier heterogeneous-agents-protocol-environment-carrier, the relation interaction-adaptation-and-outcome-relation, the record agent-types-rules-resources-actions-failures-and-oversight-held, and the boundary system-and-deployment-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:e487a5f0d5cc1e4532cc082019dc1670104767e15ea65951ca73c995830afac0; independent certificate: sha256:362a4f3bad4b21c8118a4c320bcf1ab9659d9b017152f6a1c1ba6204fb5ff239; engine receipt: sha256:60e78a62c2a13e7aca9740c5c8c924fddaf5177f7b747adcfc487785ccf4067b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

59. Collective decision

Claim: SFT-SOCIAL-COLLECTIVE-DECISION-001

Family: multi_level_causation

Exact statement: Collective decision retains eligibility, options, information, rule, participation, dissent and implementation.

Forward chain and exact carrier

The generated carrier is participants-options-rule-outcome-carrier. Its required relation is preference-input-to-decision-relation. The reconstruction must retain eligibility-options-information-rule-participation-dissent-and-time-held. Its declared empirical boundary is decision-procedure-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-MULTI-AGENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `participants-options-rule-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `decision-procedure-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `preference-input-to-decision-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `eligibility-options-information-rule-participation-dissent-and-time-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
participants-options-rule-outcome-carrier__decision-procedure-bounded__preference-input-to-decision-relation__eligibility-options-information-rule-participation-dissent-and-time-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject majority outcome as unanimous preference, truth or ethical legitimacy.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **collective decision**, the consequence is practical and exact: Collective decision retains eligibility, options, information, rule, participation, dissent and implementation. A scientific claim therefore has to expose the carrier participants-options-rule-outcome-carrier, the relation preference-input-to-decision-relation, the record eligibility-options-information-rule-participation-dissent-and-time-held, and the boundary decision-procedure-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: sha256:29f156890e5dd163e5fa6cdf8660805384bc59c0d4eb6a8721002c26320a7250; independent certificate: sha256:b135dac0f958b4c168b6806489ed99046cb56688c144e7d5ca2e803b6cf62ef1; engine receipt: sha256:ebd6ba8b976fd3197bf996059944d8bcbcad597c2e03840e9f07c0a3e356861b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

60. Aggregation and ecological inference boundary

Claim: SFT-SOCIAL-AGGREGATION-LOSS-001

Family: multi_level_causation

Exact statement: Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals.

Forward chain and exact carrier

The generated carrier is individual-records-aggregate-carrier. Its required relation is many-to-one-summary-relation. The reconstruction must retain source-rows-rule-distribution-loss-and-uncertainty-held. Its declared empirical boundary is aggregation-level-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-COLLECTIVE-MEMORY-001, SFT-SOCIAL-COLLECTIVE-DECISION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: individual-records-aggregate-carrier.
- **boundary** - which population, place, period and method bound the statement: aggregation-level-bounded.
- **relation** - which generated interaction or comparison connects the carriers: many-to-one-summary-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: source-rows-rule-distribution-loss-and-uncertainty-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

individual-records-aggregate-carrier__aggregation-level-bounded__many-to-one-summary-relation__source-rows-rule-distribution-loss-and-uncertainty-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject ecological or atomistic inference without a cross-level bridge.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- IPUMS-INTERNATIONAL-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.
- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **aggregation and ecological inference boundary**, the consequence is practical and exact: Aggregation explicitly records the many-to-one loss preventing recovery of individual states from totals. A scientific claim therefore has to expose the carrier `individual-records-aggregate-carrier`, the relation `many-to-one-summary-relation`, the record `source-rows-rule-distribution-loss-and-uncertainty-held`, and the boundary `aggregation-level-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b4bd4bd7c53a3187caef27b8737cc88c11ad9b1cc5ca3671fdd645046f88af8e; independent certificate: sha256:68db722be074104ce183ced13bcfcb9888a3d562556b9546cdcaa9fa9d631718; engine receipt: sha256:7ff9a069d479701ef5f32b29468908f4476ca0bcd77572f62d413cdd13c32765. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

61. Historical social event

Claim: SFT-SOCIAL-HISTORICAL-EVENT-001

Family: `history_intervention_ethics`

Exact statement: A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts.

Forward chain and exact carrier

The generated carrier is `actors-actions-records-context-carrier`. Its required relation is `source-to-event-reconstruction`. The reconstruction must retain `sources-custody-dates-perspectives-gaps-and-alternatives-held`. Its declared empirical boundary is `event-and-source-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-AGGREGATION-LOSS-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `actors-actions-records-context-carrier`.
- **boundary** - which population, place, period and method bound the statement: `event-and-source-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `source-to-event-reconstruction`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `sources-custody-dates-perspectives-gaps-and-alternatives-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`actors-actions-records-context-carrier__event-and-source-bounded__source-to-event-reconstruction__sources-custody-dates-perspectives-gaps-and-alternatives-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject a single archive, victor narrative or later outcome as the complete event.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **historical social event**, the consequence is practical and exact: A historical event is a source-custodied reconstruction retaining perspectives, gaps and competing accounts. A scientific claim therefore has to expose the carrier actors-actions-records-context-carrier, the relation source-to-event-reconstruction, the record sources-custody-dates-perspectives-gaps-and-alternatives-held, and the boundary event-and-source-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:249a0e97a0fdd4a27fe064ef274fb018703e8e4b79931a92c66729ff4ea4684a; independent certificate: sha256:3f26c28018ad05367627d629669f148dac685b51f74d642ff4f1058f07adc778; engine receipt: sha256:5d2c8091d96cfda76c054ea5ab140b4534508d6d7f8d2c64d102934786854533. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

62. Historical contingency

Claim: SFT-SOCIAL-CONTINGENCY-001

Family: history_intervention_ethics

Exact statement: Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals.

Forward chain and exact carrier

The generated carrier is prior-state-event-alternative-later-carrier. Its required relation is path-dependent-transition-relation. The reconstruction must retain prior-options-events-sequence-mechanisms-and-unknowns-held. Its declared empirical boundary is case-and-period-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-HISTORICAL-EVENT-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: prior-state-event-alternative-later-carrier.
- **boundary** - which population, place, period and method bound the statement: case-and-period-bounded.
- **relation** - which generated interaction or comparison connects the carriers:
path-dependent-transition-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
prior-options-events-sequence-mechanisms-and-unknowns-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
prior-state-event-alternative-later-carrier__case-and-period-bounded__path-dependent-
transition-relation__prior-options-events-sequence-mechanisms-and-unknowns-held__evid
ence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extens
ion-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject inevitability from hindsight or universal law from one historical path.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/
primary_archive_structural_correspondence. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **historical contingency**, the consequence is practical and exact: Contingency retains prior alternatives, event sequence, mechanism and unresolved counterfactuals. A scientific claim therefore has to expose the carrier
prior-state-event-alternative-later-carrier, the relation
path-dependent-transition-relation, the record
prior-options-events-sequence-mechanisms-and-unknowns-held, and the boundary
case-and-period-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 71ea865ffc8a79e0416e91d196ee64f318638ed84e66467153971cf70585c9e5; independent
certificate: sha256: 88fa6aecf2f8056955cd03d7d0aa76219c3fceff6700bd0a319d23225b32119b; engine
receipt: sha256: 79be04ee3a298f8ceca4cf7827cb30d905f125227f92f742158487be5c98a0c1. The
independent implementation recomputed the candidate product: True. All external rows were preserved: True.

63. Social causal inference

Claim: SFT-SOCIAL-CAUSAL-INFERENC-001

Family: history_intervention_ethics

Exact statement: Causal inference retains assignment, comparison, confounding, spillover, attrition and target population.

Forward chain and exact carrier

The generated carrier is units-exposure-comparison-outcome-carrier. Its required relation is counterfactual-comparison-relation. The reconstruction must retain assignment-selection-confounding-time-attrition-and-spillover-held. Its declared empirical boundary is design-and-population-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-CONTINGENCY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `units-exposure-comparison-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `design-and-population-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `counterfactual-comparison-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `assignment-selection-confounding-time-attrition-and-spillover-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
units-exposure-comparison-outcome-carrier__design-and-population-bounded__counterfact
ual-comparison-relation__assignment-selection-confounding-time-attrition-and-spillove
r-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-f
inite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject association, model coefficient or adjusted significance as causation without design support.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot sha256:cae9debd85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social causal inference**, the consequence is practical and exact: Causal inference retains assignment, comparison, confounding, spillover, attrition and target population. A scientific claim therefore has to expose the carrier units-exposure-comparison-outcome-carrier, the relation counterfactual-comparison-relation, the record assignment-selection-confounding-time-attrition-and-spillover-held, and the boundary design-and-population-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:4cd7bc322a0db065a643484da67d0ad1e4f63f66694df43856ca962b8dfdfb2f; independent certificate: sha256:0571c271a87d3c13f994c878a4f159bd8ab7da52a268df72b003b222930c0bc6; engine receipt: sha256:2ed40b2f01083326f219d9aca6f455c1d6585f185b410b751a7e4bcf18bb737e. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

64. Social intervention

Claim: SFT-SOCIAL-INTERVENTION-001

Family: history_intervention_ethics

Exact statement: Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes.

Forward chain and exact carrier

The generated carrier is population-action-implementation-outcome-carrier. Its required relation is implementation-to-outcome-comparison. The reconstruction must retain target-delivery-uptake-context-harms-and-missing-held. Its declared empirical boundary is intervention-context-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001,

SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-CAUSAL-INFERENCE-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `population-action-implementation-outcome-carrier`.
- **boundary** - which population, place, period and method bound the statement: `intervention-context-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `implementation-to-outcome-comparison`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `target-delivery-uptake-context-harms-and-missing-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
population-action-implementation-outcome-carrier__intervention-context-bounded__implementation-to-outcome-comparison__target-delivery-uptake-context-harms-and-missing-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject policy announcement as implementation or average benefit as absence of harm.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot `sha256:cae9debd85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **social intervention**, the consequence is practical and exact: Intervention evidence retains target, delivery, uptake, context, comparison, harms and missing outcomes. A scientific claim therefore has to expose the carrier `population-action-implementation-outcome-carrier`, the relation `implementation-to-outcome-comparison`, the record `target-delivery-uptake-context-harms-and-missing-held`, and the boundary `intervention-context-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal: `sha256:a392f51cd50bee633a0c092aa3877e6c3930f4a9911a3b288c624c094e7fa701`; independent certificate: `sha256:975ad9bf0c95b677a40fb54eb1eaa43198bb09847162234eec9efcb9210b71ea`; engine receipt: `sha256:aa62a0c0d10360df013c9b20d813fac4383cb3aff2e18fc51cd9e785bdbfe315`. The independent implementation recomputed the candidate product: `True`. All external rows were preserved: `True`.

65. Cross-context replication

Claim: `SFT-SOCIAL-REPLICATION-001`

Family: `history_intervention_ethics`

Exact statement: Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits.

Forward chain and exact carrier

The generated carrier is `claim-contexts-protocol-outcomes-carrier`. Its required relation is `protocol-preserving-comparison-relation`. The reconstruction must retain `populations-settings-measures-deviations-results-and-adverse-held`. Its declared empirical boundary is `transportability-bounded`. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is `SFT-FOUNDATION-ONE-001`, `SFT-FOUNDATION-FOLD-001`, `SFT-FOUNDATION-FOLD-DYNAMICS-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-INFO-QUANTITY-001`, `SFT-INFO-MUTUAL-CONDITIONAL-001`, `SFT-INFO-CHANNEL-CAPACITY-001`, `SFT-COMP-FORM-STATE-TRANSITION-001`, `SFT-COMP-DIST-CAUSALITY-001`, `SFT-COMP-DIST-COMMUNICATION-001`, `SFT-COMP-DIST-CONSENSUS-001`, `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`, `SFT-BIO-BIO-CAUSALITY-001`, `SFT-BIO-BIO-HANDOFF-001`, `SFT-CONSC-SUBJECT-CARRIER-001`, `SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001`, `SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001`, `SFT-EARTH-HAZARD-RISK-HANDOFF-001`, `SFT-ASTRO-ASTRO-HANDOFF-001`, `SFT-SOCIAL-AGGREGATION-LOSS-001`, `SFT-SOCIAL-INTERVENTION-001`. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `claim-contexts-protocol-outcomes-carrier`.
- **boundary** - which population, place, period and method bound the statement: `transportability-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `protocol-preserving-comparison-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `populations-settings-measures-deviations-results-and-adverse-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

`claim-contexts-protocol-outcomes-carrier__transportability-bounded__protocol-preserving-comparison-relation__populations-settings-measures-deviations-results-and-adverse-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule`

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the

result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject same label with changed construct as replication or one failure as universal impossibility.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot `sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5`.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot `sha256:cae9debda85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c`.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **cross-context replication**, the consequence is practical and exact: Replication retains populations, settings, measures, protocol deviations, adverse outcomes and transport limits. A scientific claim therefore has to expose the carrier `claim-contexts-protocol-outcomes-carrier`, the relation `protocol-preserving-comparison-relation`, the record `populations-settings-measures-deviations-results-and-adverse-held`, and the boundary `transportability-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:18a70a4476b3ee35b194b7f8ccfdd667c0074be162501095795a9fe3e9ddf70e; independent certificate: sha256:f8ac24f95b26a32d2dbb87825021f55b3b97ae7b0057f45eb2338bc42a9fa227; engine receipt: sha256:dd32b12f5464a8ce1964f7e13f600dcc4394a95554cee8702add49c94872de4a. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

66. Ethical and value boundary

Claim: SFT-SOCIAL-ETHICS-BOUNDARY-001

Family: history_intervention_ethics

Exact statement: Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law.

Forward chain and exact carrier

The generated carrier is affected-agents-values-actions-consequences-carrier. Its required relation is normative-deliberation-and-rights-relation. The reconstruction must retain stakeholders-values-rights-harms-benefits-disagreement-and-power-held. Its declared empirical boundary is ethical-decision-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-AGGREGATION-LOSS-001, SFT-SOCIAL-REPLICATION-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: affected-agents-values-actions-consequences-carrier.
- **boundary** - which population, place, period and method bound the statement: ethical-decision-bounded.
- **relation** - which generated interaction or comparison connects the carriers: normative-deliberation-and-rights-relation.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: stakeholders-values-rights-harms-benefits-disagreement-and-power-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
affected-agents-values-actions-consequences-carrier__ethical-decision-bounded__normative-deliberation-and-rights-relation__stakeholders-values-rights-harms-benefits-disagreement-and-power-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject empirical frequency as moral permission or moral commitment as measured causal fact.

- `false_premise` - passed True: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed True: reject a changed registered source identity
- `tampered_artifact` - passed True: reject a missing, duplicate or additional survivor
- `boundary` - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 2. Unresolved transports retained: 1. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- UCDP-DOWNLOADS-001 - transport captured; captured bytes 138078; snapshot
sha256:5b8c5d85582971fe6f30b161886704b38b241b12cd18fea29ea3d762f8e5ebc5.
- VDEM-DATA-001 - transport captured; captured bytes 122924; snapshot
sha256:cae9debd85e6c4008ea246ae2f1f38c915172fb18b087e303f803913026bb0c.
- ICPSR-DATA-001 - transport failed_preserved; captured bytes 0; snapshot unresolved and preserved.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **ethical and value boundary**, the consequence is practical and exact: Ethical judgment retains affected agents, rights, harms, benefits, disagreement and power and is not relabelled an empirical law. A scientific claim therefore has to expose the carrier affected-agents-values-actions-consequences-carrier, the relation normative-deliberation-and-rights-relation, the record stakeholders-values-rights-harms-benefits-disagreement-and-power-held, and the boundary ethical-decision-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256: 72308cfc9208ff4581dda71edfb062eacb81d0853d418deff376c5bf05591140; independent certificate: sha256: 0571a291a9639095487b59a34be87ecc6ae2a260f849c24db5e1384771bc278f; engine receipt: sha256: 3cfb0486415cdab2c5ce460390aab1782820d675af4b08e782e175442c5e13b3. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

67. Mathematics and graph handoff

Claim: SFT-SOCIAL-MATH-HANDOFF-001

Family: ownership_handoffs

Exact statement: Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity.

Forward chain and exact carrier

The generated carrier is social-carrier-abstract-structure-carrier. Its required relation is typed-abstraction-handoff. The reconstruction must retain social-meaning-measurement-map-loss-and-use-held. Its declared empirical boundary is disciplinary-ownership-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-carrier-abstract-structure-carrier`.
- **boundary** - which population, place, period and method bound the statement: `disciplinary-ownership-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `typed-abstraction-handoff`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `social-meaning-measurement-map-loss-and-use-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
social-carrier-abstract-structure-carrier__disciplinary-ownership-bounded__typed-abst
raction-handoff__social-meaning-measurement-map-loss-and-use-held__evidence-and-norma
tive-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-al
l-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject an unlabeled graph as a social network law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdedb725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **mathematics and graph handoff**, the consequence is practical and exact: Social Science supplies measured meaning; Mathematics supplies abstract structure without selecting social identity. A scientific claim therefore has to expose the carrier social-carrier-abstract-structure-carrier, the relation typed-abstraction-handoff, the record social-meaning-measurement-map-loss-and-use-held, and the boundary disciplinary-ownership-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:ce19e3e32dc03520abda5709fab08478bb7bd746922a600f5eef94fc1a3a6862; independent certificate: sha256:b07d808e6f7948ffe4823a000f88871dfd85b650814ad747b692095a42a67c7d; engine receipt: sha256:fcdfc6fbd20780ccaf6c91aef1fc4f054fc9f6357e3612bc6d6d7a869cd5b599b. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

68. Computation and distributed-system handoff

Claim: SFT-SOCIAL-COMPUTATION-HANDOFF-001

Family: ownership_handoffs

Exact statement: Computation may simulate a declared social process but cannot make simulated behavior an observation.

Forward chain and exact carrier

The generated carrier is social-process-computational-model-carrier. Its required relation is semantics-preserving-handoff. The reconstruction must retain agents-rules-assumptions-model-errors-and-use-held. Its declared empirical boundary is model-purpose-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001,

SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-MATH-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-process-computational-model-carrier`.
- **boundary** - which population, place, period and method bound the statement: `model-purpose-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `semantics-preserving-handoff`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `agents-rules-assumptions-model-errors-and-use-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
social-process-computational-model-carrier__model-purpose-bounded__semantics-preserving-handoff__agents-rules-assumptions-model-errors-and-use-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject platform behaviour or consensus protocol as a universal human law.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **computation and distributed-system handoff**, the consequence is practical and exact: Computation may simulate a declared social process but cannot make simulated behaviour an observation. A scientific claim therefore has to expose the carrier social-process-computational-model-carrier, the relation semantics-preserving-handoff, the record agents-rules-assumptions-model-errors-and-use-held, and the boundary model-purpose-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:200da5093ee18b4dc8a953ef6eee664db6ed30f644e64e6cce62b119d2d6b2da; independent certificate: sha256:3d21e8532e7bcd4b0323620a1d943ee8c462e61782ffaafd3471bd8830bd367b7; engine receipt: sha256:de0cc7b5d201335a2cfa310fc1bc03d706361bd52e26a6eee479e7e16b30a210. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

69. Biology, consciousness and individual cognition handoff

Claim: SFT-SOCIAL-BIO-CONSC-HANDOFF-001

Family: ownership_handoffs

Exact statement: Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations.

Forward chain and exact carrier

The generated carrier is individual-organism-mind-social-context-carrier. Its required relation is cross-level-evidence-handoff. The reconstruction must retain mechanism-experience-behaviour-context-and-limit-held. Its declared empirical boundary is level-of-analysis-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-COMPUTATION-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: individual-organism-mind-social-context-carrier.
- **boundary** - which population, place, period and method bound the statement: level-of-analysis-bounded.
- **relation** - which generated interaction or comparison connects the carriers: cross-level-evidence-handoff.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: mechanism-experience-behaviour-context-and-limit-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

individual-organism-mind-social-context-carrier__level-of-analysis-bounded__cross-level-evidence-handoff__mechanism-experience-behaviour-context-and-limit-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report,

model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject biological reduction of an institution or social attribution of a private quale without evidence.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **biology, consciousness and individual cognition handoff**, the consequence is practical and exact: Individual mechanisms and experience remain upstream evidence linked to, but not replaced by, collective relations. A scientific claim therefore has to expose the carrier `individual-organism-mind-social-context-carrier`, the relation `cross-level-evidence-handoff`, the record `mechanism-experience-behaviour-context-and-limit-held`, and the boundary `level-of-analysis-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b352745a8a219c3f808ce3340b1808bcb4821181e6326f935dd5012cc208542a; independent certificate: sha256:739b84d64a33d4a526397434e53adf10cb1680e635fc48c971288064521e374d; engine receipt: sha256:7a26afef3725b313ba83e780dc752a863110a77f0e03c86e9d2e6ae177eaac0f. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

70. Medicine and public-health handoff

Claim: SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001

Family: ownership_handoffs

Exact statement: Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both.

Forward chain and exact carrier

The generated carrier is health-state-population-social-condition-carrier. Its required relation is individual-population-and-social-link. The reconstruction must retain clinical-measures-exposure-access-distribution-and-context-held. Its declared empirical boundary is health-ownership-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-BIO-CONSC-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: health-state-population-social-condition-carrier.
- **boundary** - which population, place, period and method bound the statement: health-ownership-bounded.
- **relation** - which generated interaction or comparison connects the carriers:
individual-population-and-social-link.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible:
clinical-measures-exposure-access-distribution-and-context-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct:
evidence-and-normative-class-explicit.

- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer:
root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case:
positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
health-state-population-social-condition-carrier__health-ownership-bounded__individual-population-and-social-link__clinical-measures-exposure-access-distribution-and-context-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject population association as individual diagnosis or clinical efficacy as equal access.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **medicine and public-health handoff**, the consequence is practical and exact: Medicine owns clinical state; Social Science owns collective access and distribution; public-health composition retains both. A scientific claim therefore has to expose the carrier health-state-population-social-condition-carrier, the relation individual-population-and-social-link, the record clinical-measures-exposure-access-distribution-and-context-held, and the boundary health-ownership-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:b2c21f28a49ac9c99765eb89ac4d244115474b0e53d80e2117071aed2c2a0383; independent certificate: sha256:69afd5d99d267834a5ac23272274a4d9c612b005b6241aa8bed5a9d8f2dadf7a; engine receipt: sha256:2f5748560880bdbbc3ad54e5dd3457d3df4707505ed574df7d6b319f7cc29595. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

71. Earth and environmental handoff

Claim: SFT-SOCIAL-EARTH-HANDOFF-001

Family: ownership_handoffs

Exact statement: Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact.

Forward chain and exact carrier

The generated carrier is hazard-population-exposure-impact-carrier. Its required relation is environment-to-social-impact-chain. The reconstruction must retain hazard-exposure-vulnerability-response-time-and-ownership-held. Its declared empirical boundary is hazard-impact-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-MEDICINE-PUBLIC-HEALTH-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: hazard-population-exposure-impact-carrier.
- **boundary** - which population, place, period and method bound the statement: hazard-impact-bounded.
- **relation** - which generated interaction or comparison connects the carriers: environment-to-social-impact-chain.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: hazard-exposure-vulnerability-response-time-and-ownership-held.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: evidence-and-normative-class-explicit.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: root-bound-forward-forcing.
- **generality** - whether finite extension preserves every context rather than universalising one case: positive-finite-extension-retains-all-contexts.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: no-extra-rule.

Their ordered product gives the unique survivor:

```
hazard-population-exposure-impact-carrier__hazard-impact-bounded__environment-to-social-impact-chain__hazard-exposure-vulnerability-response-time-and-ownership-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is depth_independent. Minimality passed: True. Named-form uniqueness passed: True. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject physical hazard as identical social harm or policy as an Earth law.

- false_premise - passed True: reject the generated form lacking the complete physical carrier
- tampered_source - passed True: reject a changed registered source identity
- tampered_artifact - passed True: reject a missing, duplicate or additional survivor
- boundary - passed True: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: True. External evidence selected the survivor: false. Credential or prestige was used as evidence: false. Consensus was used as proof: false. Normative judgment was relabelled observation: false.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdeb725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **earth and environmental handoff**, the consequence is practical and exact: Earth Science owns hazard; Social Science owns exposure, vulnerability, collective response and unequal impact. A scientific claim therefore has to expose the carrier hazard-population-exposure-impact-carrier, the relation environment-to-social-impact-chain, the record hazard-exposure-vulnerability-response-time-and-ownership-held, and the boundary hazard-impact-bounded together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:12fac511becdbef67e5d84123fe6d053e9b23ba153f6ea3954daa7c96dc01ad7; independent certificate: sha256:fddb59eb2a127570a7094b446dfc5ec0040231e6347ce305e8eb5380d7ee641b; engine receipt: sha256:949369f3908bb753e62f2b5017113fc3d1b5ab7d7d62ab6c26366ce55a227aa0. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

72. Engineering and policy implementation handoff

Claim: SFT-SOCIAL-ENGINEERING-HANDOFF-001

Family: ownership_handoffs

Exact statement: Engineering may test implementations against social laws but applications cannot select or rewrite those laws.

Forward chain and exact carrier

The generated carrier is social-law-design-deployment-consequence-carrier. Its required relation is law-to-implementation-test-relation. The reconstruction must retain requirements-design-users-context-failures-harms-and-feedback-held. Its declared empirical boundary is deployment-purpose-bounded. These coordinates are not labels added after a result: they were part of the sealed obligation that generated the candidate space.

The dependency chain is SFT-FOUNDATION-ONE-001, SFT-FOUNDATION-FOLD-001, SFT-FOUNDATION-FOLD-DYNAMICS-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-ORDER-LATTICE-001, SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-OPTIMIZATION-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-QUANTITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-DIST-CAUSALITY-001, SFT-COMP-DIST-COMMUNICATION-001, SFT-COMP-DIST-CONSENSUS-001, SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001, SFT-BIO-BIO-CAUSALITY-001, SFT-BIO-BIO-HANDOFF-001, SFT-CONSC-SUBJECT-CARRIER-001, SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001, SFT-MED-CLINICAL-EVIDENCE-HANDOFF-001, SFT-EARTH-HAZARD-RISK-HANDOFF-001, SFT-ASTRO-ASTRO-HANDOFF-001, SFT-SOCIAL-ETHICS-BOUNDARY-001, SFT-SOCIAL-EARTH-HANDOFF-001. Each dependency was model-admitted before this claim. The present claim adds no axiom, free parameter or fitted rule and cannot use an external target, credential, consensus position or prior answer to select its form.

Generation rule: Generate the literal Cartesian product of eight preregistered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external source identities or outcomes are opened; filter only by registered preservation properties.

Grammar boundary: Exactly eight binary dimensions: 256 forms. Structural closure is depth-independent for positive finite contextual extension; empirical correspondence remains population, place, period, method and protocol bounded.

Complete enumeration and uniqueness

The eight binary axes generate exactly 256 literal candidates. Their preserving coordinates are:

- **carrier** - who or what bears the social state: `social-law-design-deployment-consequence-carrier`.
- **boundary** - which population, place, period and method bound the statement: `deployment-purpose-bounded`.
- **relation** - which generated interaction or comparison connects the carriers: `law-to-implementation-test-relation`.
- **record** - which identities, alternatives, minorities, failures and missing rows remain reconstructible: `requirements-design-users-context-failures-harms-and-feedback-held`.
- **evidence** - whether observation, report, archive, inference, model and normative judgment remain distinct: `evidence-and-normative-class-explicit`.
- **provenance** - whether the result traces to the One instead of prestige, consensus or a prior answer: `root-bound-forward-forcing`.
- **generality** - whether finite extension preserves every context rather than universalising one case: `positive-finite-extension-retains-all-contexts`.
- **extension** - whether any fitted weight, exception or opaque oracle has been added: `no-extra-rule`.

Their ordered product gives the unique survivor:

```
social-law-design-deployment-consequence-carrier__deployment-purpose-bounded__law-to-implementation-test-relation__requirements-design-users-context-failures-harms-and-feedback-held__evidence-and-normative-class-explicit__root-bound-forward-forcing__positive-finite-extension-retains-all-contexts__no-extra-rule
```

Each of the other 255 forms differs on at least one registered coordinate. Erasing the carrier loses who or what bears the claim. Erasing the boundary universalises a sample or period. Importing the relation allows a target or status hierarchy to choose the result. Retaining only a favourable aggregate removes minority and failed states. Conflating evidence classes turns a report, model or value judgment into an observation. Breaking provenance removes the trace to the One. Erasing finite context makes one case universal. Adding an exception or opaque oracle creates a free rule. Because every non-survivor commits at least one such loss and the survivor commits none, uniqueness is enumerated rather than asserted.

The closure certificate is `depth_independent`. Minimality passed: `True`. Named-form uniqueness passed: `True`. The base case retains the least positive finite carrier, relation, context, record and evidence class. The successor adds one lawful agent, relation, group, context, period or record while preserving all earlier identities, adverse rows and source boundaries.

Falsification and deliberately unfavourable controls

The registered falsification condition is: Reject a successful product metric as scientific admission or a failed implementation as automatic law falsification.

- `false_premise` - passed `True`: reject the generated form lacking the complete physical carrier
- `tampered_source` - passed `True`: reject a changed registered source identity
- `tampered_artifact` - passed `True`: reject a missing, duplicate or additional survivor
- `boundary` - passed `True`: reject target access, forbidden proof values, imported laws and free parameters

A passed control means the altered condition was rejected as required; it is not a reward for an adverse scientific outcome. If a premise, source, artifact or boundary change were accepted, admission would halt. Automation generated repeated records from the frozen specification but did not select the survivor, change a condition or decide that a failure had passed.

Post-seal external evidence

Evidence class: `source_bound_observation_survey_archive_or_documented_record/primary_archive_structural_correspondence`. Captured registered sources: 3. Unresolved transports retained: 0. The registered structural consequence corresponded: `True`. External evidence selected the survivor: `false`. Credential or prestige was used as evidence: `false`. Consensus was used as proof: `false`. Normative judgment was relabelled observation: `false`.

- WHO-GHO-001 - transport captured; captured bytes 141734; snapshot
sha256:84fba66a22eeb625b3b75885b352fb035563767a9933e96045452ff32286d76f.
- CROSSREF-RECORDS-001 - transport captured; captured bytes 69898; snapshot
sha256:32ceaf90269046a7bf55fdebd725d304909afac6e76dda8e1016a6d83f3898ba.
- WORLD-BANK-WDI-001 - transport captured; captured bytes 1874098; snapshot
sha256:4bc9c30cf5a69fb3f6736831fa5d4aef9f9958c48877a48ac3a65fc500a37927.

The comparison is structural and source-bound. It checks that a relevant primary archive can represent the declared carrier, population or institutional context, period, provenance and missingness without being allowed to rewrite the Fold law. It does not claim that one archive exhausts every society, that a dataset category is metaphysically final, or that a source custodian's reputation proves the result. New records can extend or challenge the empirical surface only with their source, population, period, method and adverse rows retained.

Scientific meaning

For **engineering and policy implementation handoff**, the consequence is practical and exact: Engineering may test implementations against social laws but applications cannot select or rewrite those laws. A scientific claim therefore has to expose the carrier `social-law-design-deployment-consequence-carrier`, the relation `law-to-implementation-test-relation`, the record `requirements-design-users-context-failures-harms-and-feedback-held`, and the boundary `deployment-purpose-bounded` together. If any one is hidden, the result may remain a description, interpretation, model, political commitment or application choice, but it is not this admitted law.

This distinction protects both people and science. It prevents an aggregate from erasing the individuals who compose it, a dominant record from erasing those excluded from its archive, a successful outcome from erasing harms and failures, and institutional authority from deciding whether a derivation is true. It also prevents criticism alone from becoming admission: a replacement must still generate its forms, eliminate alternatives, survive controls, preserve source custody and reproduce through the shared engine.

Machine certificate

Candidate count: 256; survivor count: 1; derivation seal:

sha256:bb77400f2642664067344c2a6f2c5e24a581e91925ee2042c03bdb3f7d02da35; independent certificate: sha256:246a3d6fc5d8f22e35a4eda6cb7151641ee9b197ec890e9faf82022b27775c0e; engine receipt: sha256:bfe9b4bbad15adb6a0d7bac2472d99f035c9ac7810b25c365d92b6f5b74b1f87. The independent implementation recomputed the candidate product: True. All external rows were preserved: True.

Historical reconciliation

The inherited V1/V2 census contains 763 entries. Every entry was classified atomically by present branch ownership. Five Social-owned questions-collective recurrence, emergence, synchronization and multi-agent composition-were registered as questions and mapped to present claims. Their former answers were not imported as premises. Graph, distributed-computation, cognition, medicine, Earth and engineering questions remain typed handoffs rather than being relabelled Social laws.

What this foundation does and does not claim

It claims that the presently known foundational forms required to state and test Social and Collective Sciences have been generated, uniquely closed, independently reconstructed and source-bound at their declared boundaries. It does not claim that every historical event is known, every archive is accessible, every society shares one parameter, every ethical dispute is solved by frequency, or all future discovery is closed. It does not turn institutions into persons, collectives into private conscious subjects, simulations into observations or applications into law selectors.

The branch is complete to Maria Smith's current standard and evidence, and open to lawful extension. A new discovery may add a carrier, relation, source, context, historical record or stronger bridge. It must preserve existing adverse, absent and failed records and pass the same engine. No branch is permanently locked against valid additions.

Full-field continuation roadmap

- complete social network group organisation and institution census.
- complete anthropology archaeology sociology family and community studies.
- complete governance law politics economics labour and inequality evidence.
- complete demography migration geography city and settlement evidence.
- complete culture language education religion media and knowledge transmission.
- complete criminology justice conflict peace and social-harm evidence.
- complete multi-agent artificial collective and institutional knowledge studies.
- complete causal intervention replication ethics and cross-context limits.

Every later extension will be decomposed into nonduplicate obligations, sealed before outcomes, admitted claim by claim and incorporated through a versioned paper update. Applications, public policy and engineered platforms may test the laws but may not select them.

Reproduction and present validation state

During branch construction the repository used proportionate checks: both immutable seals around each admission, one 256-form census and independent reconstruction per claim, and the six Social-focused tests after integration. After all fifteen foundations were integrated, the untouched repository command `python3 -m sft verify-all` passed on 28 July 2026: 1,319/1,319 derivations replayed, 969/969 unit and end-to-end tests passed, 1,264/1,264 core executable lines covered, 1,011 empirical claims audited, 114/114 formal Physics claims reaching measurement and 5/5 live exact NIST/CODATA checks. The measured runtime was 4,753.740 seconds.

This branch passed 72 sequential admissions, 72 independent validators, 18,432 candidate decisions, its integration audit and six focused tests. Neither the engine nor verification authority was modified. The paper and evidence release are published open access at DOI [10.5281/zenodo.21640814](https://doi.org/10.5281/zenodo.21640814).

Rights, participation and scientific admission

Anyone may read, reproduce, criticize and extend the work. No credential is required to speak. The Ernos Labs designation requires the shared empirical constitution: visible root derivation, exact grammar, complete enumeration, no target selection, no hidden fit, preserved unfavourable evidence, independent reconstruction, unchanged protected authority, retained authorship and open licensing. Criticism is public participation; admission is a separately evidenced result.

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GitHub: <https://github.com/MettaMazza>

Paper license: CC BY 4.0

Code license: Apache-2.0

Machine identities

- Inventory: sha256:8b5f0e4c65aa077ab8dad91c3cb00f466068ee387d4f54e63d3df57d56e1745c
- Prior audit: sha256:38d9c04161cffe5f852094afb223a3affc3fbf6776cc89acc86b472c1a7465dc
- Integration audit:
sha256:315798d5cebf6e65fb8d0a436eaeef8171d3386599bd44896f1cff910905a86699
- Source registry: sha256:5ecee0b682729015412e612c6dd71f599dbcb1c4c6a3276b6ac230beab7037af
- External target census:
sha256:f82d4757217304bc64b8210b5f82bc0d301091c38ce358b83d95d460bcfbdf3c
- Engine: sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification authority:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8

Complete current-census successor supplement

The authoritative current scope of this paper is 76 live model-admitted claims (social_collective 4, social_collective_systems 72). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 4 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Exact preferential-flow inequality

Claim ID: SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002

Exact statement: Repeated preferential routing generates complementary exact shares while conserving One.

Dependency route: SFT-SOCIAL-DISTRIBUTION-001 -> SFT-SOCIAL-POWER-001 ->
SFT-MATH-DYNAMICAL-SYSTEMS-001

Candidate generator: Generate complete eight-axis product for SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002 and independently reconstruct the witness.

Grammar boundary: All positive finite binary allocation depths with reversed, equal and fitted controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-parts-and-labels__binary-retained-share-and-complement__declare
d-context__complete-agent-institution-time-record__complete-product__after-seal__favo
rable-adverse-broken-reversed-unresolved__no-extra-rule

Exact result: At depth n the shares are $1/2^n$ and One minus $1/2^n$. They complete One at every step and the retained majority rises strictly across every positive finite successor.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:04a60e9ee095d1a90f1a9802b18ab38a4a98ee3ef7c5072d87db95aead474896 at receipts/engine/model_admitted/SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002-04a60e9ee095d1a9.json

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Consensus/polarization Fold lock

Claim ID: SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002

Exact statement: A common half-One image cannot distinguish consensus from retained polarized preimages.

Dependency route: SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002 ->
SFT-SOCIAL-COLLECTIVE-DECISION-001 -> SFT-COMP-DIST-CONSENSUS-001 ->
SFT-INFO-CONSERVATION-LOSS-001

Candidate generator: Generate complete eight-axis product for SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002 and independently reconstruct the witness.

Grammar boundary: The complete two-fibre lock with label-retained and label-lost observation classes.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-parts-and-labels__distinct-quarter-three-quarter-preimages-common-half-image__declared-context__complete-agent-institution-time-record__complete-product__after-seal__favorable-adverse-broken-reversed-unresolved__no-extra-rule

Exact result: Quarter-One and three-quarter-One are distinct polarized carriers, both Fold to half-One and together complete One. Consensus requires closure of retained labels; the common image alone cannot prove agreement.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:02d9e32e8d89c68b88668db175e2d8de4cdacba0f3adaef2a40b3fe2b0d3ee56 at receipts/engine/model_admitted/SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002-02d9e32e8d89c68b.json

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Dissipative social period-two cycle

Claim ID: SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002

Exact statement: The least nontrivial recurrent collective carrier alternates one-third and two-thirds and never rests at its balance.

Dependency route: SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002 -> SFT-SOCIAL-SOCIAL-CYCLE-001 -> SFT-SOCIAL-INSTITUTIONAL-CHANGE-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001

Candidate generator: Generate complete eight-axis product for SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002 and independently reconstruct the witness.

Grammar boundary: The least positive period-two orbit with shocks, broken-cycle and context controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-parts-and-labels__one-third-two-thirds-closed-cycle__declared-context__complete-agent-institution-time-record__complete-product__after-seal__favorable-adverse-broken-reversed-unresolved__no-extra-rule

Exact result: One-third Folds to two-thirds and two-thirds to one-third. The states complete One, their balance is half-One, and the orbit never reaches or rests at One. Boom/bust or political labels are context-bound correspondences.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:fcfeb4b70b7ccc725e116368b85180508377d5cad166860b6a1c72d88a666d07 at receipts/engine/model_admitted/SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002-fcfeb4b70b7ccc72.json

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Complete post-seal exact Social comparison

Claim ID: SFT-SOCIAL-VALIDATION-EXACT-COMPLETE-FAMILY-002

Exact statement: The sealed three-law family is compared across separately registered inequality, polarization/consensus and cycle observations with full context retention.

Dependency route: SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002 -> SFT-SOCIAL-CAUSAL-INFERERENCE-001 -> SFT-SOCIAL-CONTINGENCY-001 -> SFT-PHYS-MEAS-TARGET-CUSTODY-001

Candidate generator: Generate complete eight-axis product for SFT-SOCIAL-VALIDATION-EXACT-COMPLETE-FAMILY-002 and independently reconstruct the witness.

Grammar boundary: The sealed family against identity-first cross-context sources.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-parts-and-labels__sealed-social-family-versus-cross-context-observations__declared-context__complete-agent-institution-time-record__complete-product__after-seal__favorable-adverse-broken-reversed-unresolved__no-extra-rule

Exact result: The comparison tests cumulative preferential advantage, polarization hidden by aggregation and recurrent collective cycles while retaining interventions, broken cycles, reversals, institutional dependence and unresolved causation.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: empirically_tested_and_independently_replicated

Engine receipt: sha256:6ee94ca2079585d1d8cb8eb995bed82c5a4206bd2ef3d2eca1e2c61fc838744c at receipts/engine/model_admitted/SFT-SOCIAL-VALIDATION-EXACT-COMPLETE-FAMILY-002-6ee94ca2079585d1.json

Control	Passed	Expected	Observed
false_premise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.0.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

Social science fails when a convenient aggregate is allowed to erase its people, when an institutional label stands in for an observed process, when prestige becomes evidence, when dissent becomes error by declaration, or when an ethical preference is hidden inside an empirical claim. This paper reconstructs the foundational Social and Collective Sciences branch of Smithian Fold Theory from the One through admitted mathematical, informational, computational, biological, consciousness, medical, Earth and astronomical dependencies. The result comprises twelve families and seventy-two laws governing agents, groups, population, observation, communication, coordination, conflict, institutions, trust, knowledge, exchange, distribution, power, networks, culture, multi-level causation, history, intervention, ethics and disciplinary handoff. The proof layer adds no axiom, free parameter, fit, conventional numerical zero, negative proof scalar, irrational or imaginary proof number, completed infinity or ungenerated continuum. Every claim is a literal 256-form census across eight binary axes. External records are opened only after the complete 18,432-candidate branch seal. Each result is then independently reconstructed and compared against source-custodied evidence while adverse, unavailable and unresolved records remain visible. The paper gives the full chain for every claim and establishes an extension-open empirical constitution for collective inquiry.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Data and code availability

The successor Markdown is publications/successors/social_collective_systems/FROM_ONE_RELATION_TO_SOCIETY_PAPER_001_V1_1.md and its scientific predecessor is publications/current/social_collective_systems/FROM_ONE_RELATION_TO_SOCIETY.md. The Lean sources and report are under generated/lean4_validation/; the report is generated/lean4_validation/reports/whole_model_validation.json. Current claims are indexed by census/claims.json; complete candidates, decisions, controls, certificates, observations and receipts remain under claims/<claim-id>/ and receipts/. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

References and source registry

1. Smith, Maria. The preceding version of this paper, preserved at the repository path identified above, 2026.
2. Smith, Maria. *There Is No Nothing*. Ernos Labs Methods Paper 00, versioned publication series, 2026.
3. de Moura, Leonardo, and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language." *Automated Deduction - CADE 28*, 2021. DOI: [10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
4. Lean FRO. *The Lean Language Reference*. lean-lang.org/doc/reference/latest.
5. Ernos Labs. Current claim packages, source registries and Lean verification report, repository snapshot bound by the machine identities below.

Successor machine-identity appendix

- Preceding source:
publications/current/social_collective_systems/FROM_ONE_RELATION_TO_SOCIETY.md
- Preceding source SHA-256:
sha256:83105339ba4dd4f419f06935ff7926264510e62588f78bc58bdd79d1788c9ef8
- Lean report: generated/lean4_validation/reports/whole_model_validation.json
- Lean report SHA-256:
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf
- Ordered census SHA-256:
sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71
- Execution manifest SHA-256:
sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41
- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned PASS. Its immutable report identity for this successor is

sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **76** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.1.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 76 current claim(s) within `social_collective_systems`, `social_collective` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.